./whoami

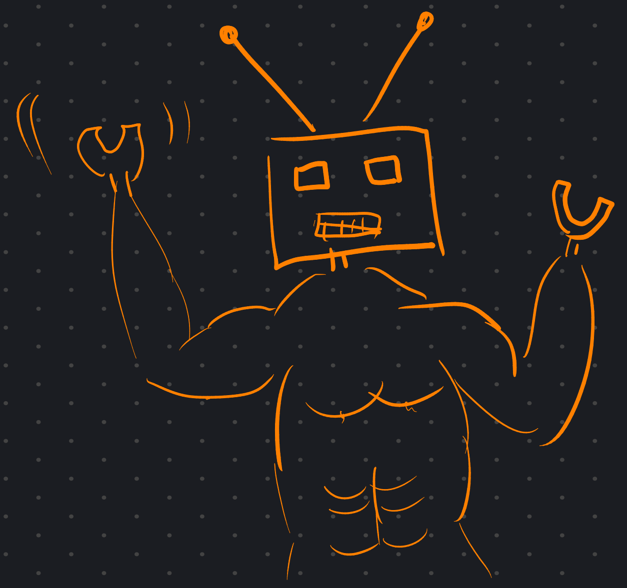
GIORGIO STRANO

PhD at GLADIA

TRANSFORMER NERD

RESEARCH IN AI SAFETY, MECHANISTIC INTERPRETABILITY

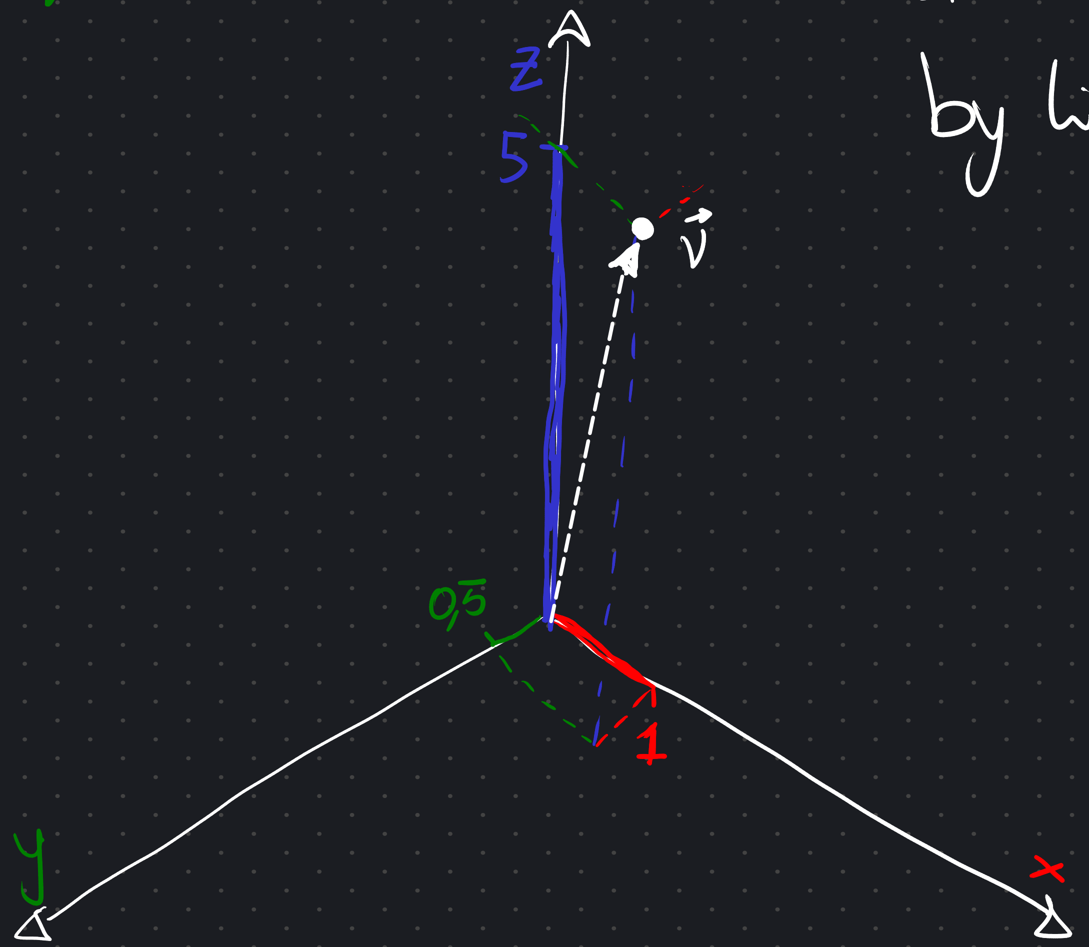
1. Recap on Transformers



This is a 3D vector.

$$\vec{v} = \begin{array}{|c|c|c|} \hline 1 & 0.5 & 5 \\ \hline \multicolumn{1}{c}{x} & \multicolumn{1}{c}{y} & \multicolumn{1}{c}{z} \end{array}$$

It represents a POINT in 3D SPACE by listing its COORDINATES.

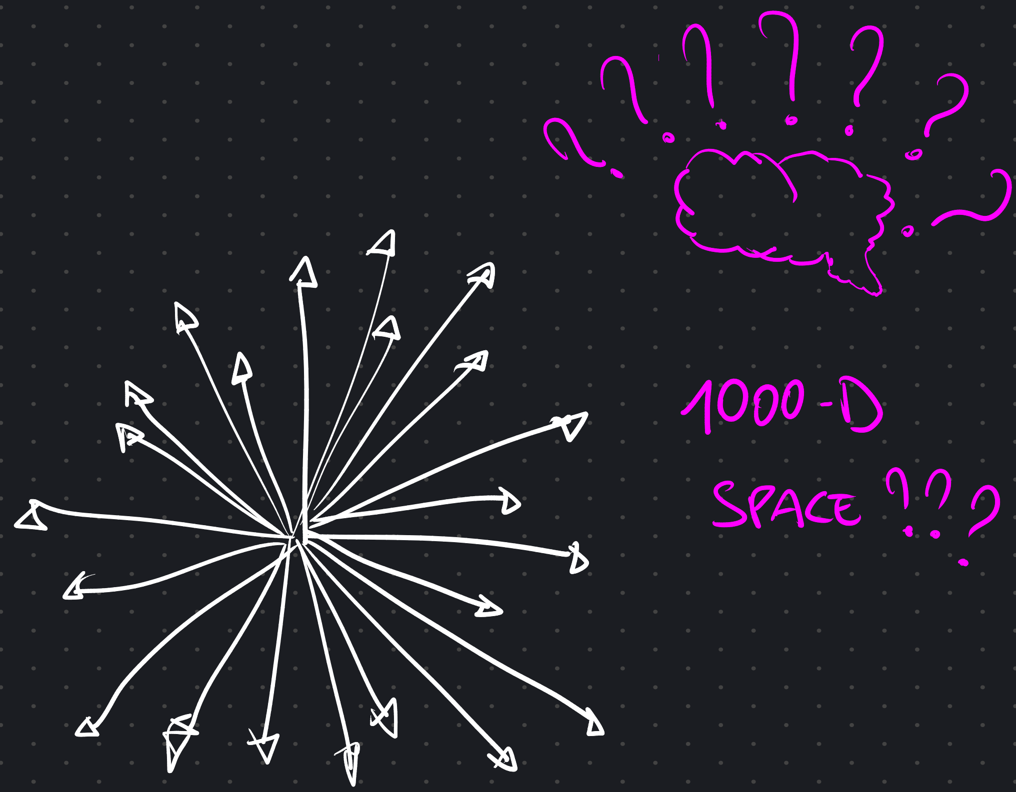


This is a 1000 dimensional vector.

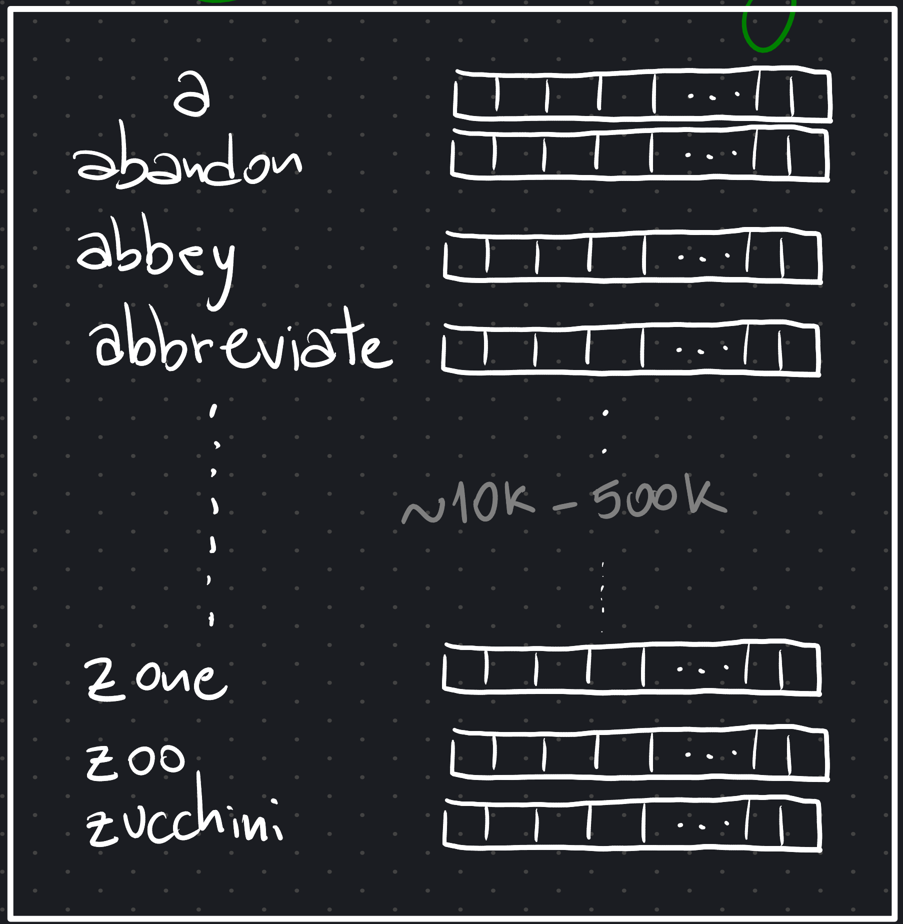
$$\underbrace{[\,|\,|\,|\,|\,\cdots\,|\,|\,|\,|\,]}_{\text{1000 numbers}} \in \mathbb{R}^{1000}$$

It represents the COORDINATES of a POINT

in a 1000-DIMENSIONAL space

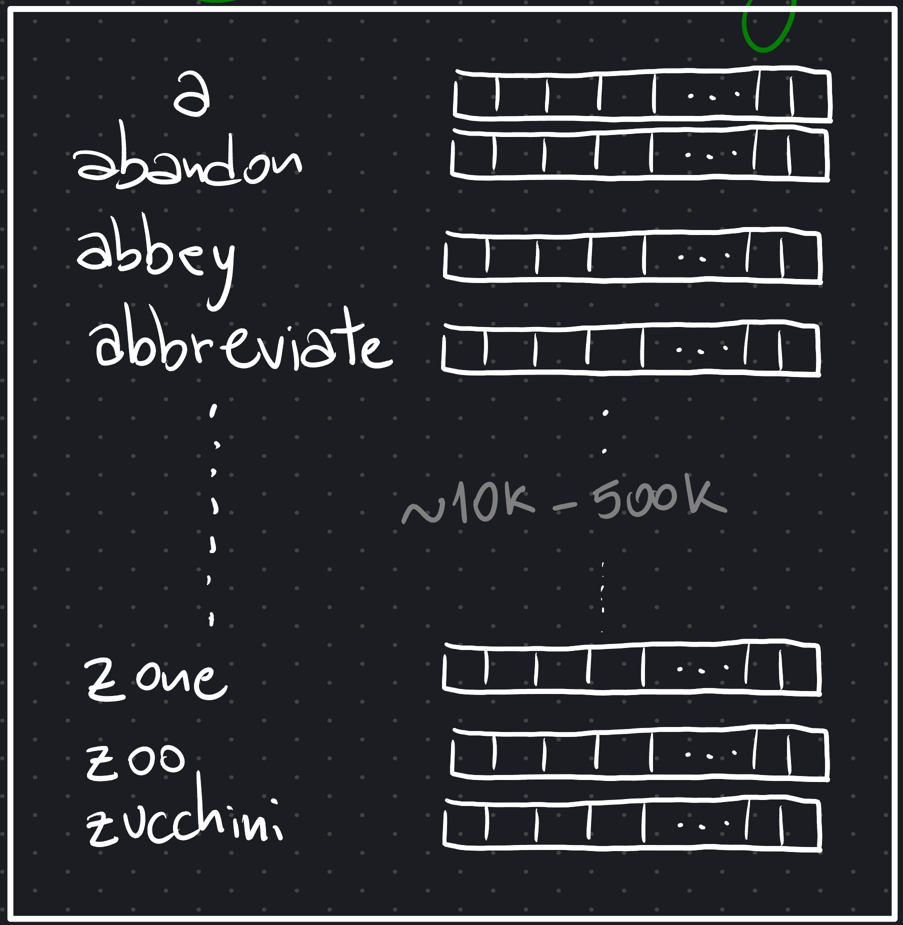


The Dictionary



During TRAINING, the model creates a DICTIONARY, assingning ONE VECTOR to EACH TOKEN

The Dictionary



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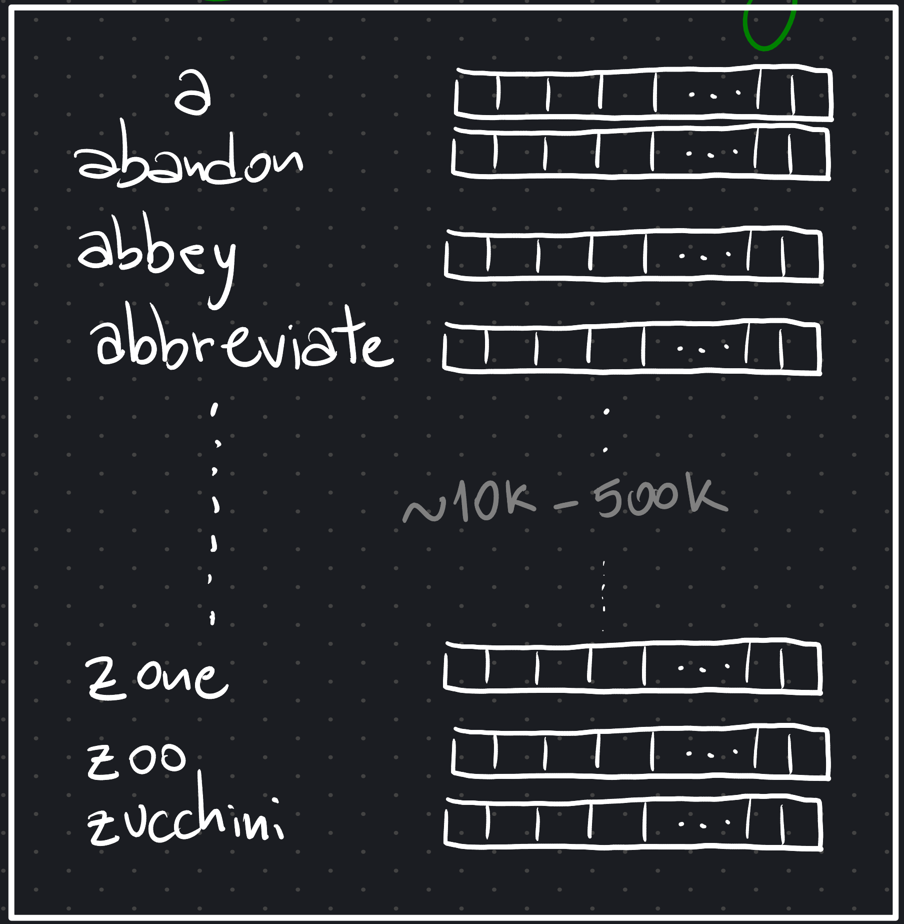
These VECTORS are called EMBEDDINGS

E(pizza) = 

The Dictionary

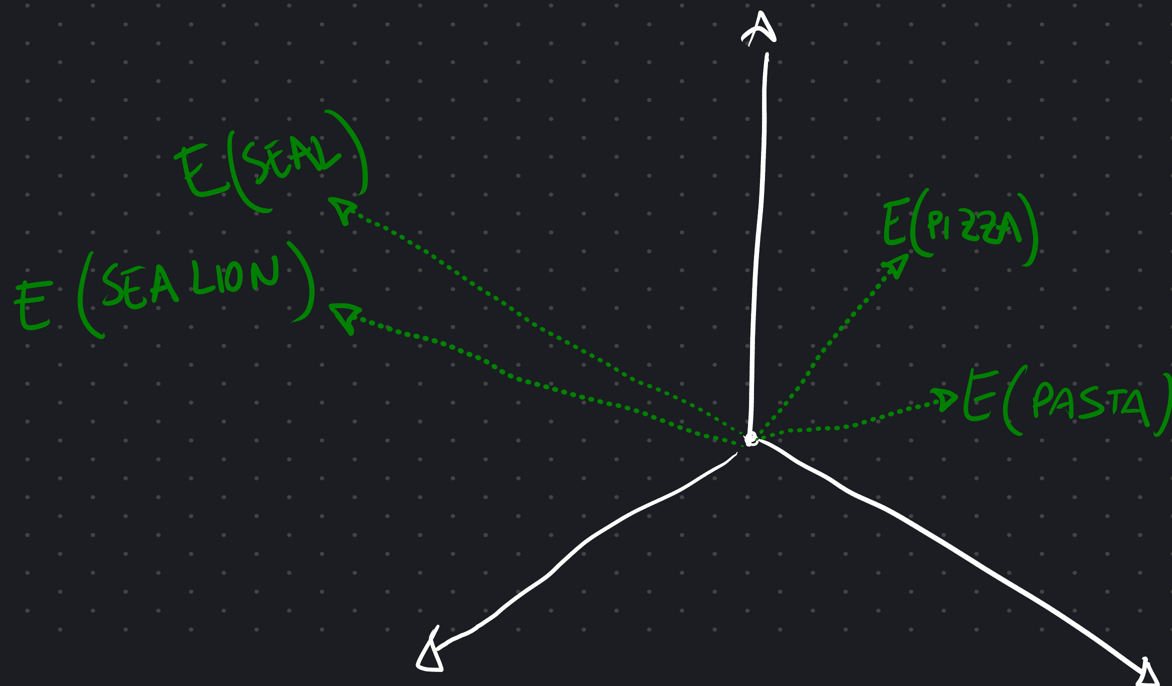


During TRAINING, the model creates a DICTIONARY, assigning ONE VECTOR to EACH TOKEN

These VECTORS are called **EMBEDDINGS**

E(pizza) =

IF we had EMBEDDINGS with only 3 DIMENSIONS, we could PLOT them in 3D space



EMBEDDINGS have some very COOL PROPERTIES.

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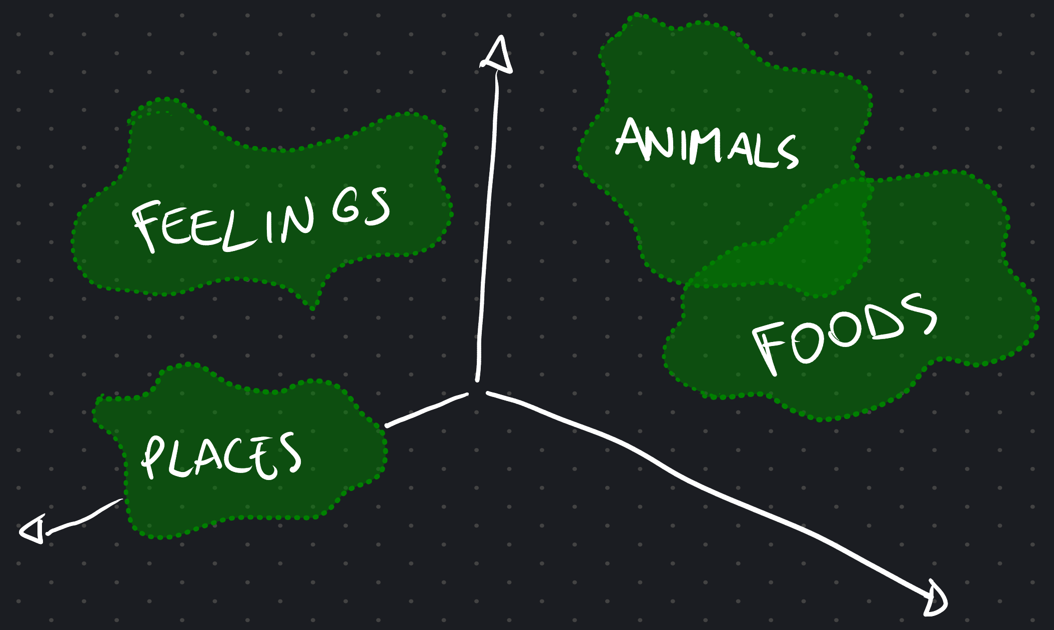
1. They are GROUPED in space SEMANTICALLY

$E(\text{CAT}) \approx E(\text{FELINE})$



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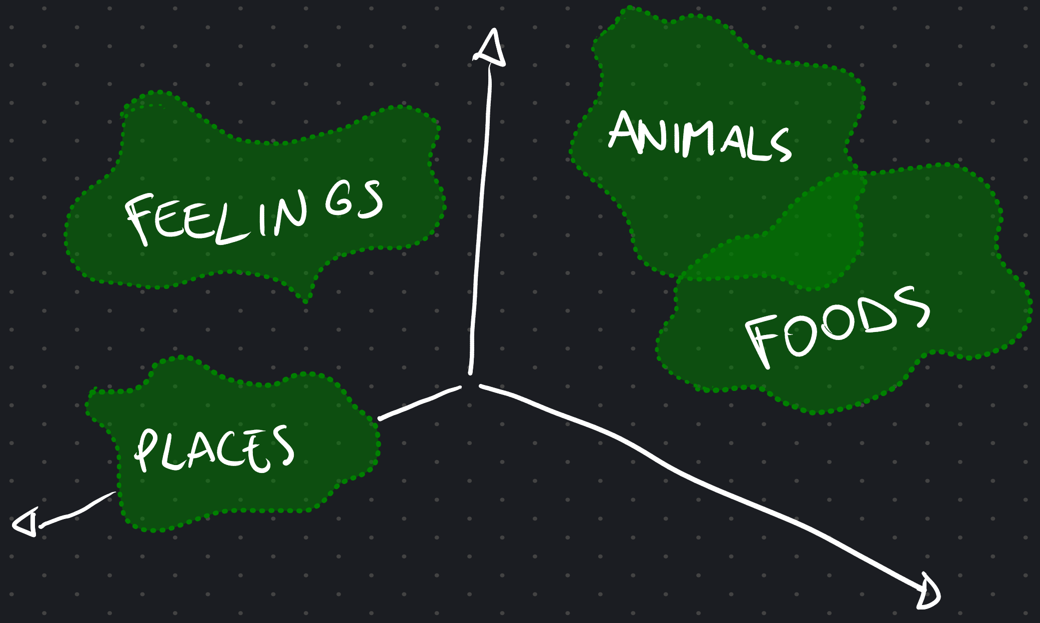
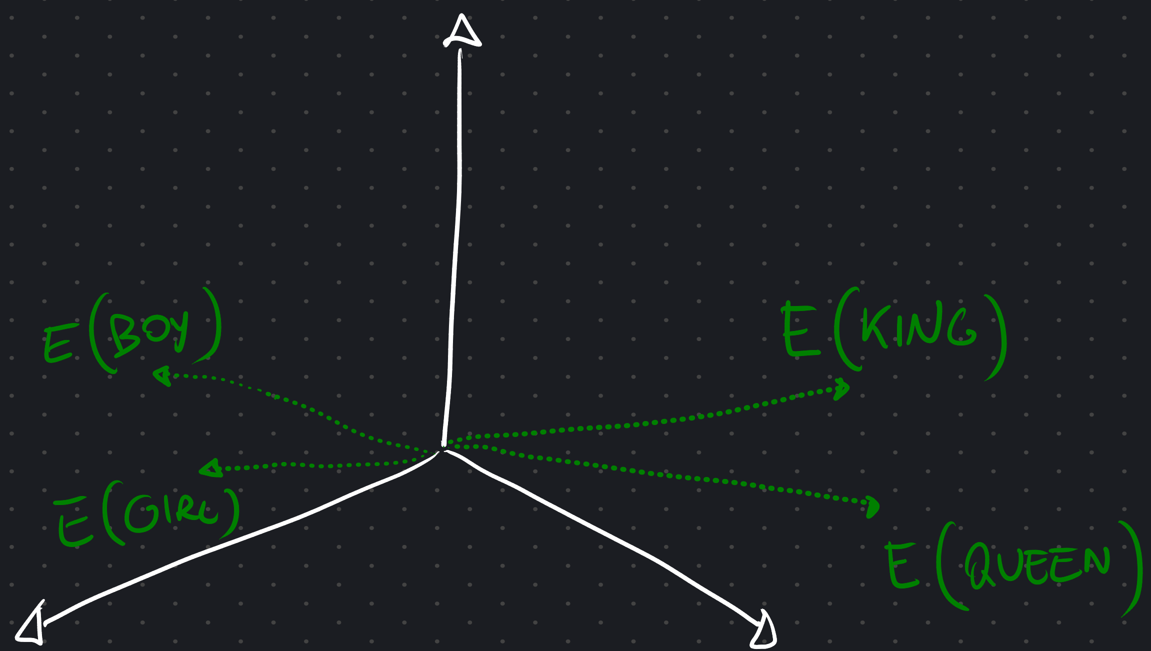


2. DIRECTIONS encode MEANING



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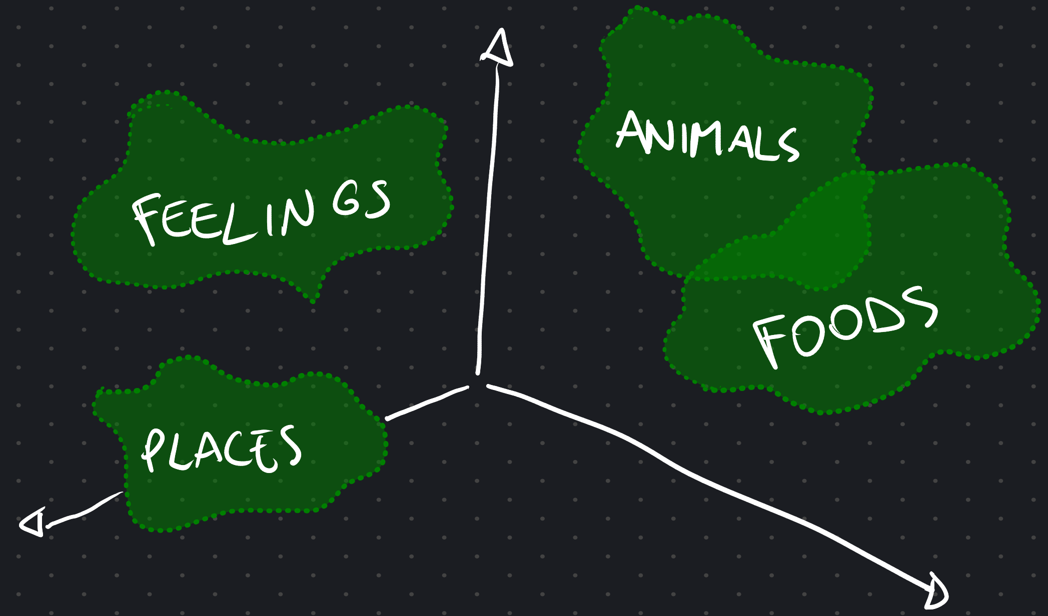
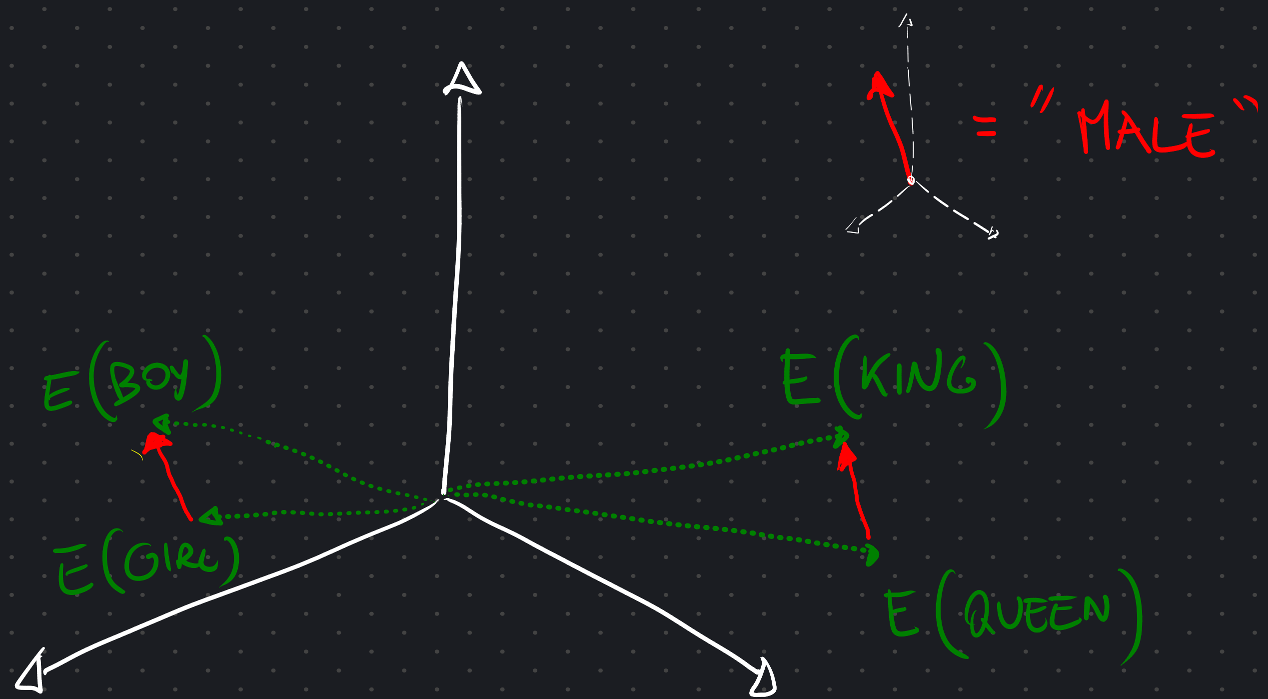
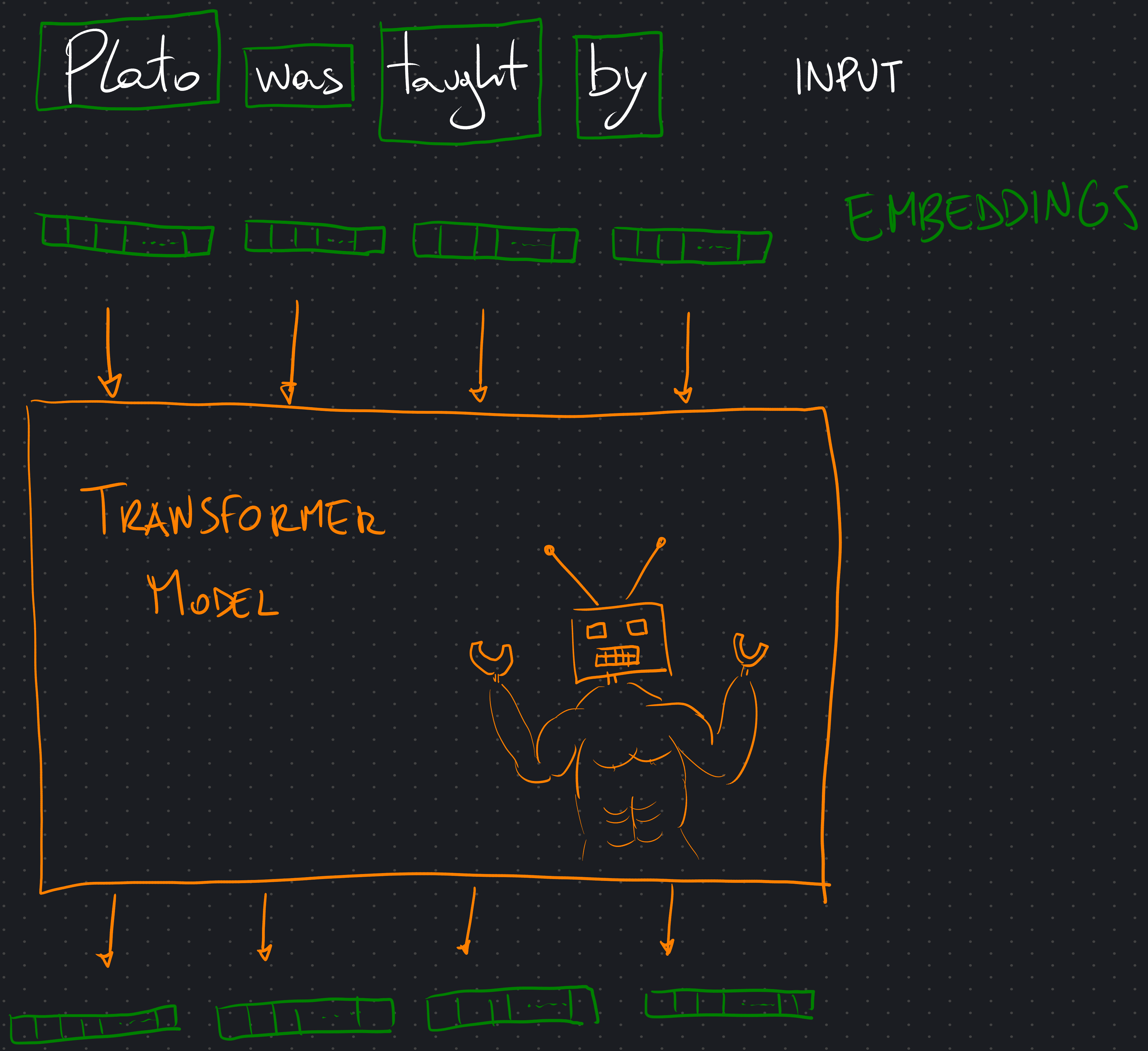


2. DIRECTIONS encode MEANING

$$E(\text{KING}) - E(\text{BOY}) + E(\text{GIRL}) \approx E(\text{QUEEN})$$



TRANSFORMERS ARE SEQUENCE-TO-SEQUENCE MODELS



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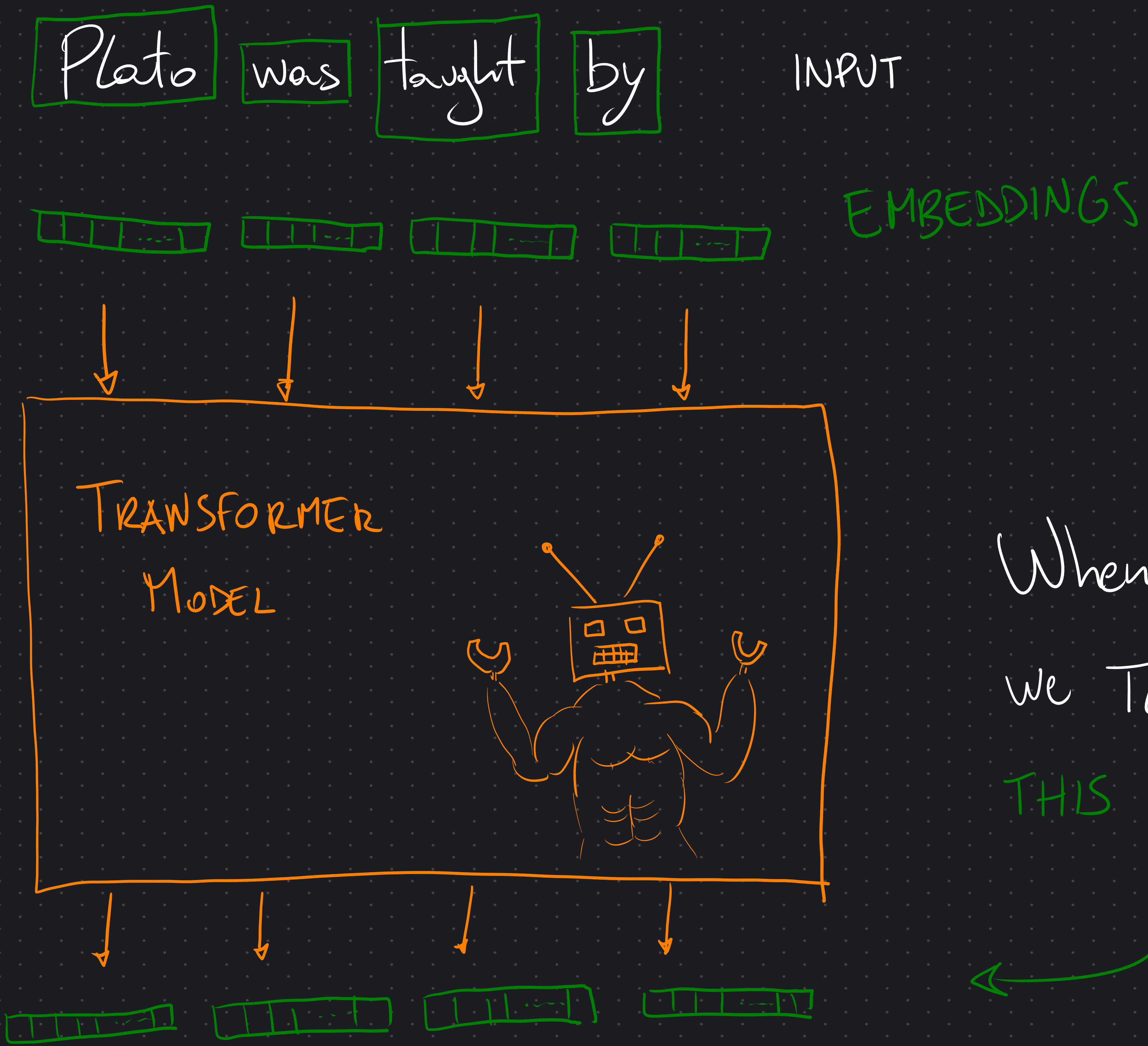
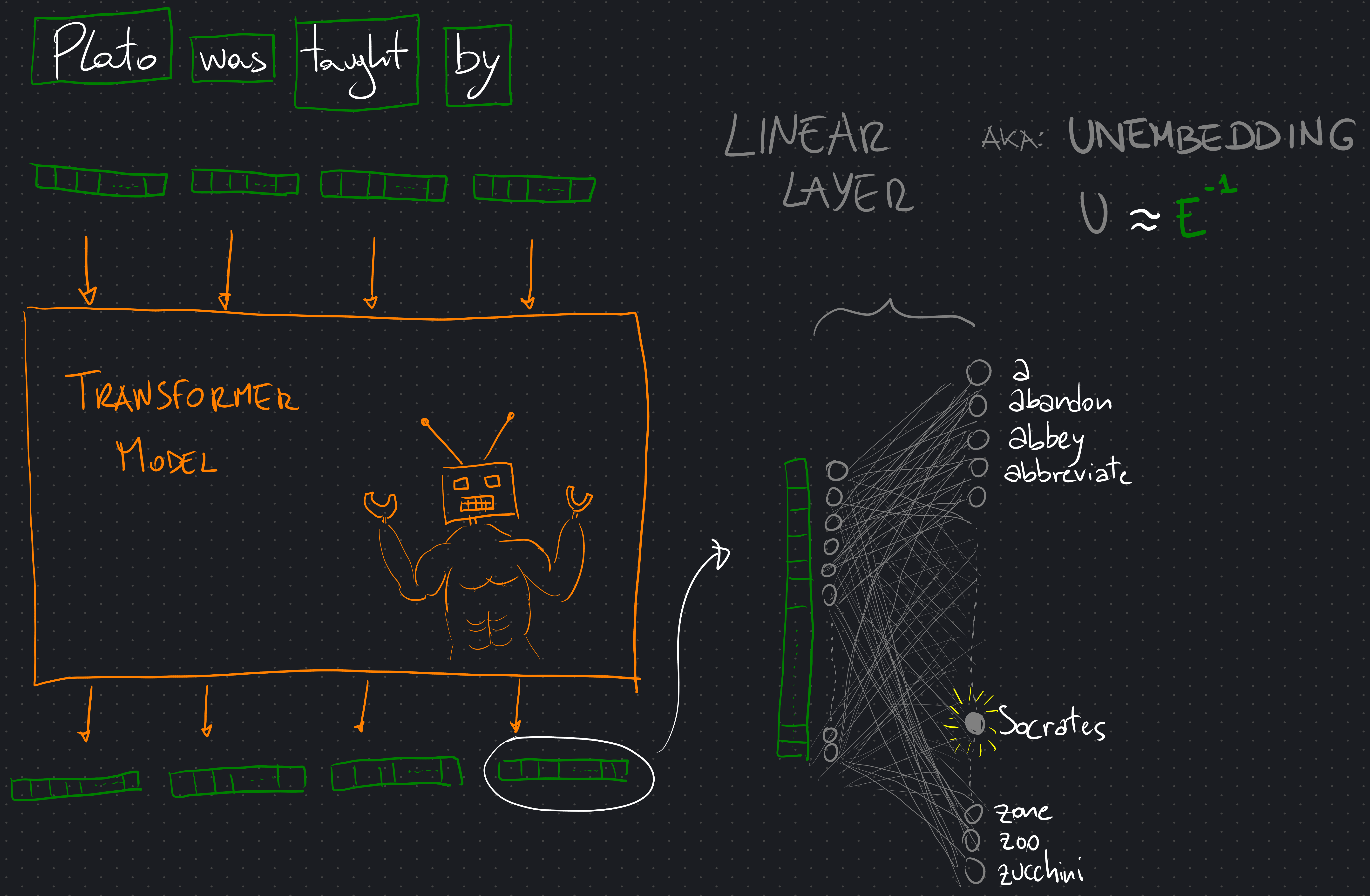
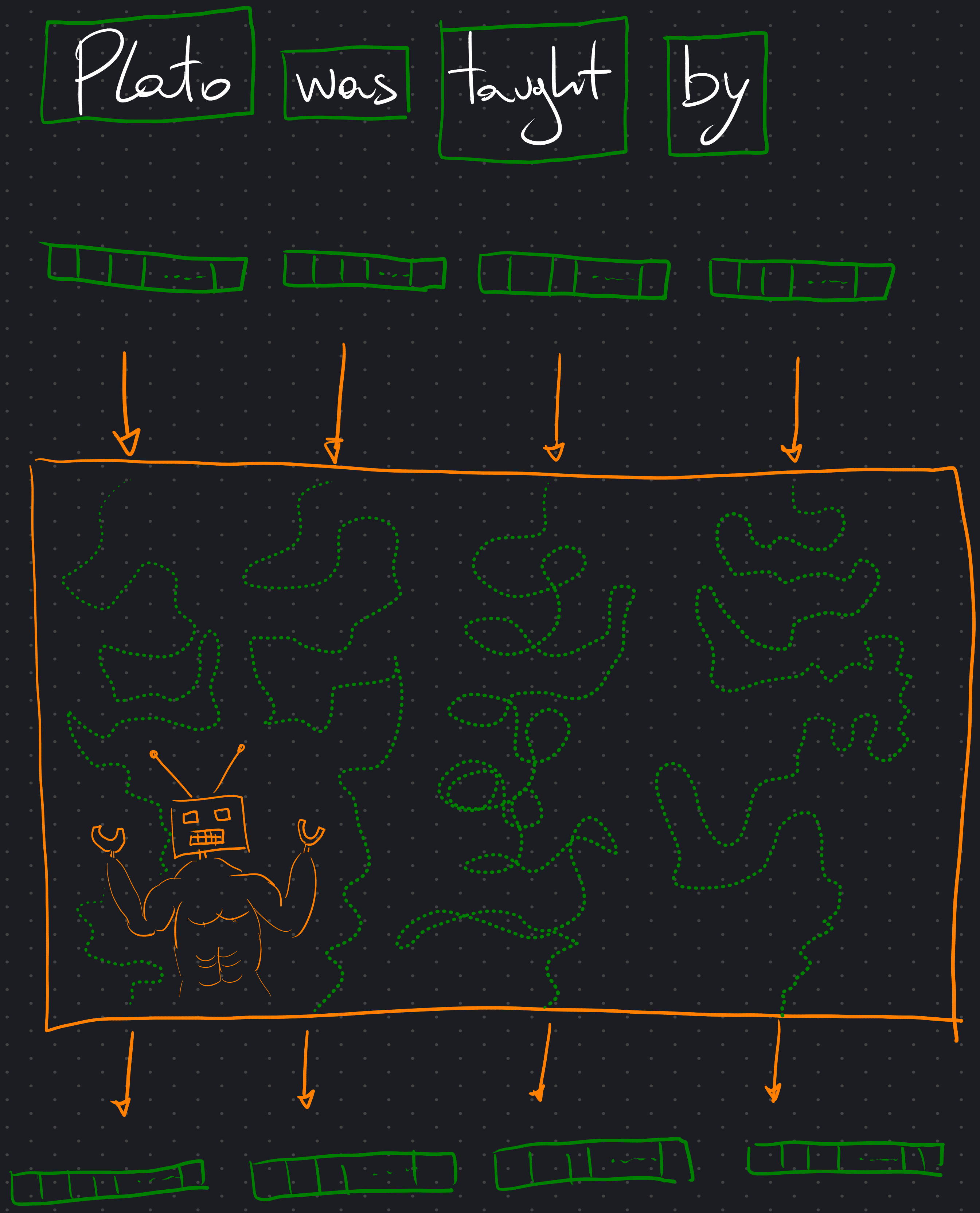


When we TRAIN them, we TELL them WHAT THIS SEQUENCE NEEDS TO BE.

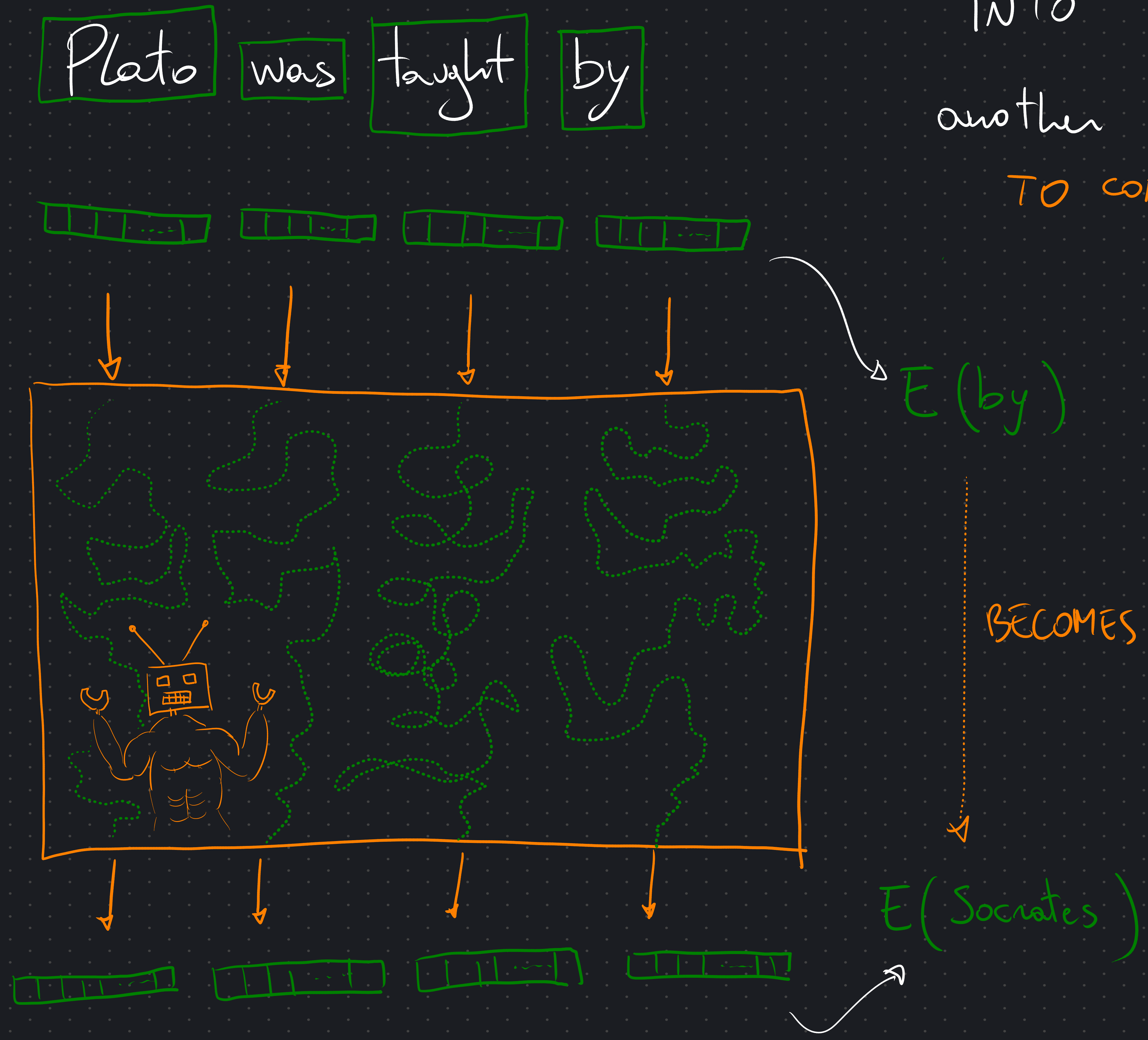
LANGUAGE MODELING



So, the model TRANSFORMS each EMEDDING of a WORD



So, the model TRANSFORMS each EMEDDING of a WORD INTO the EMBEDDING of another WORD that is LIKELY TO COME NEXT.



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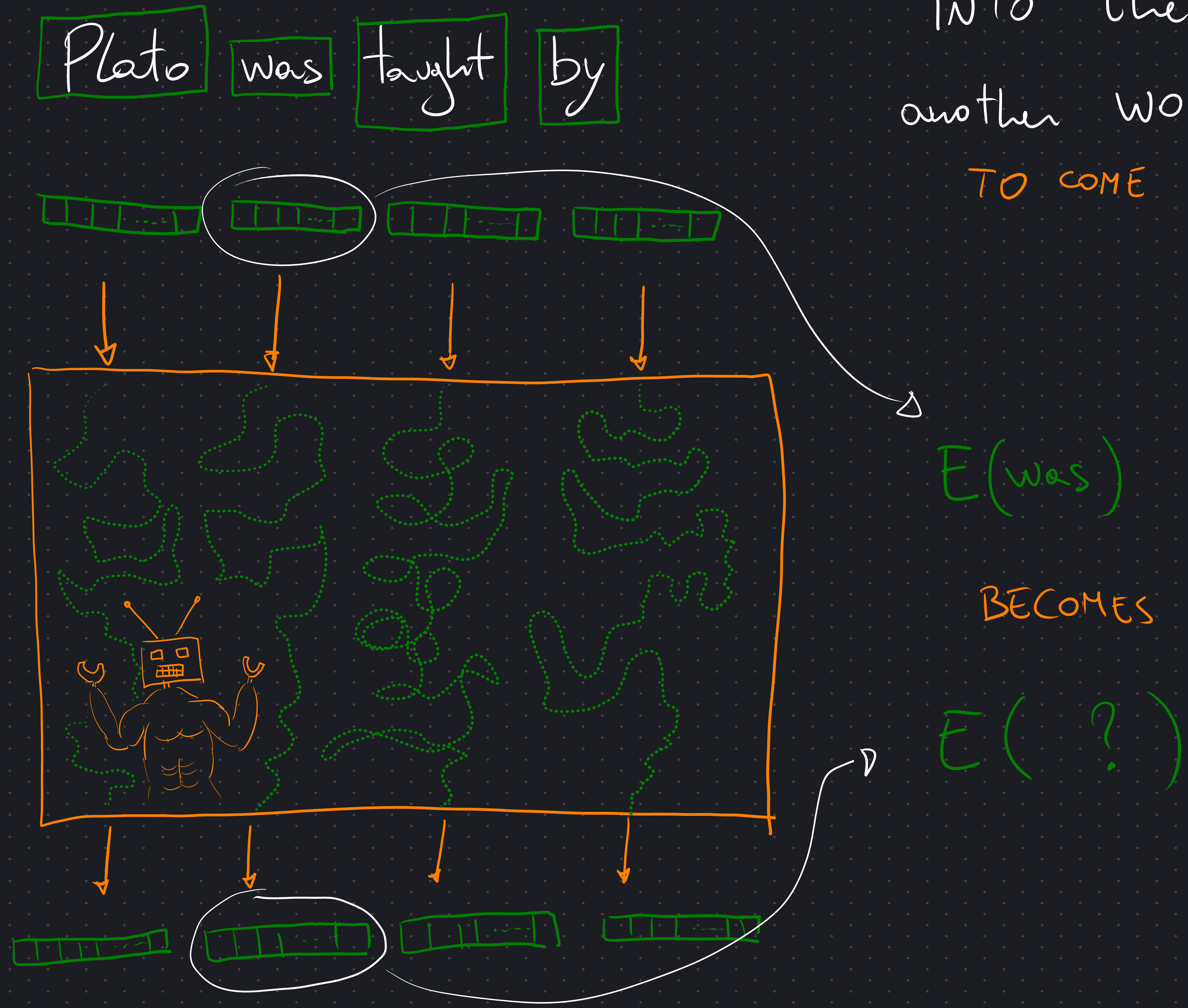

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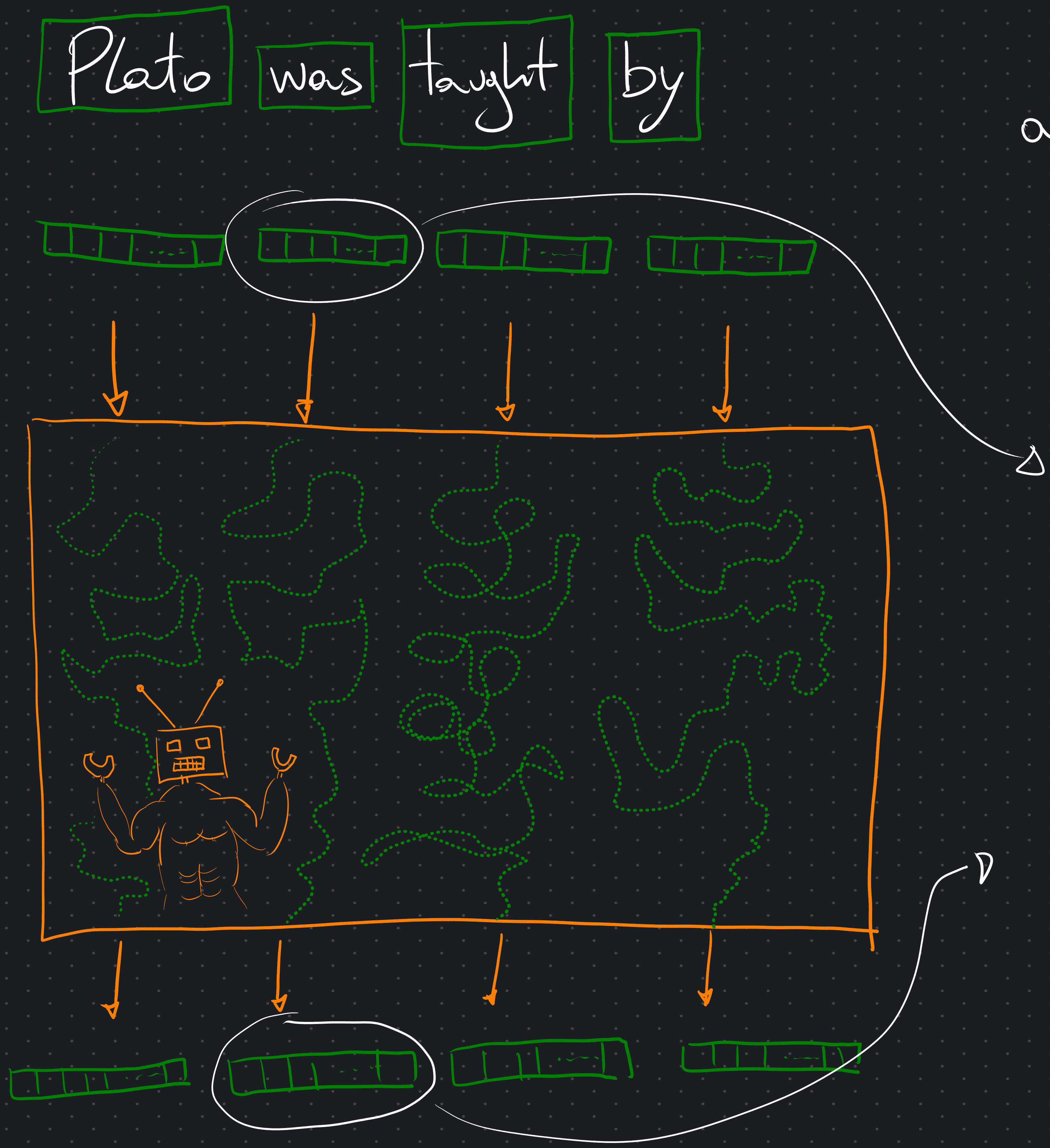


E(was)

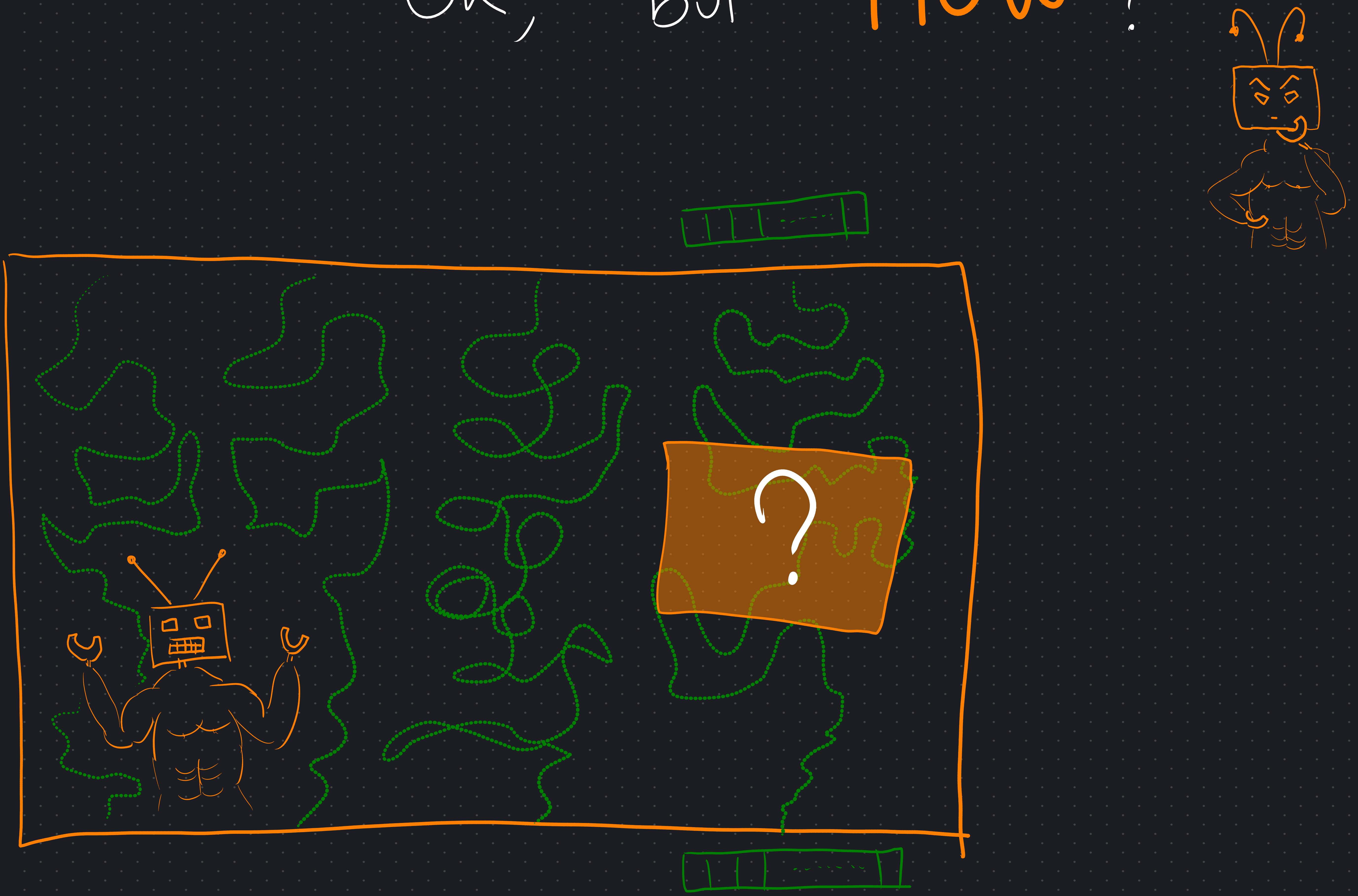
BECOMES

E(a)

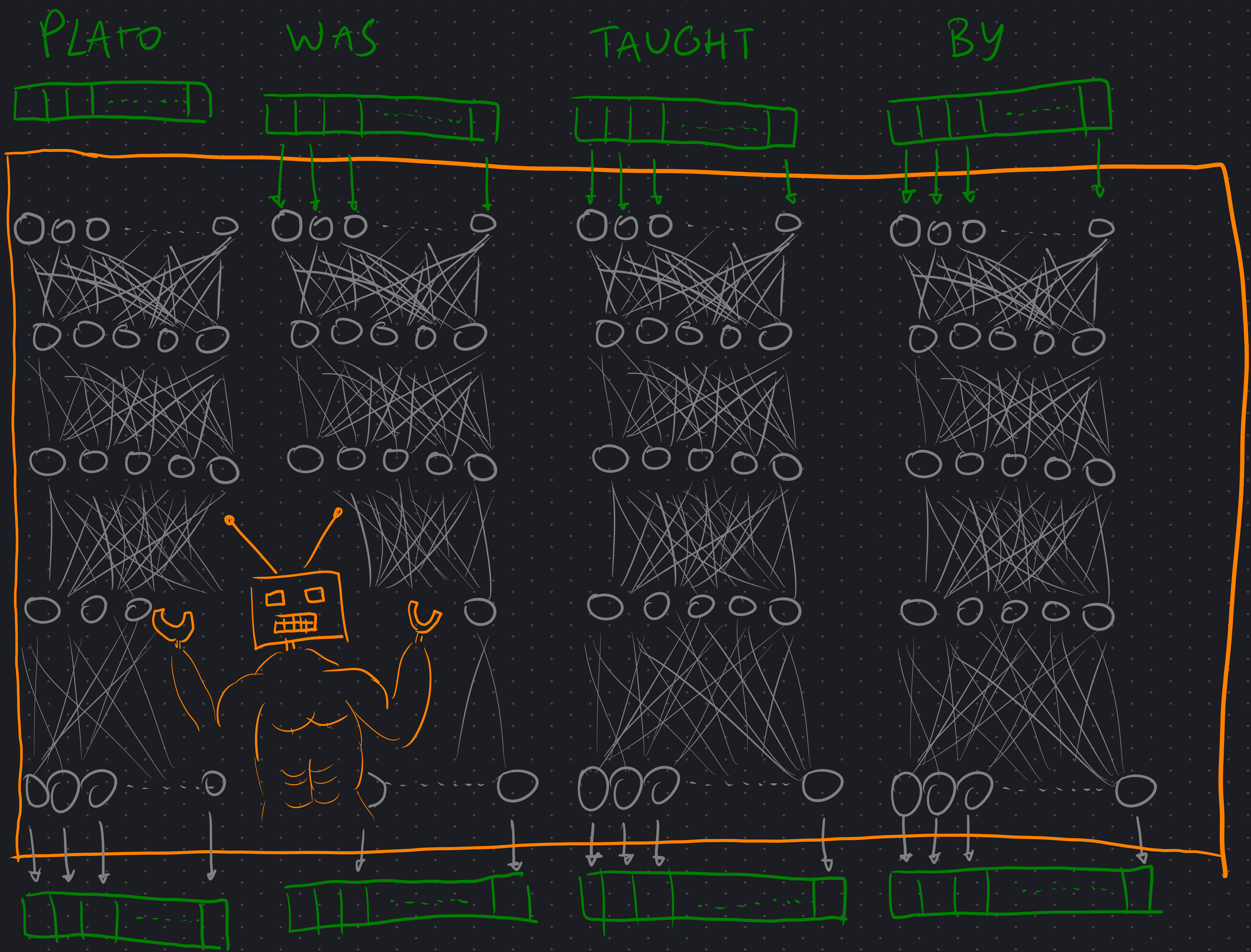
MODEL is thinking:

"Plato was a philosopher..."

OK, but HOW ?



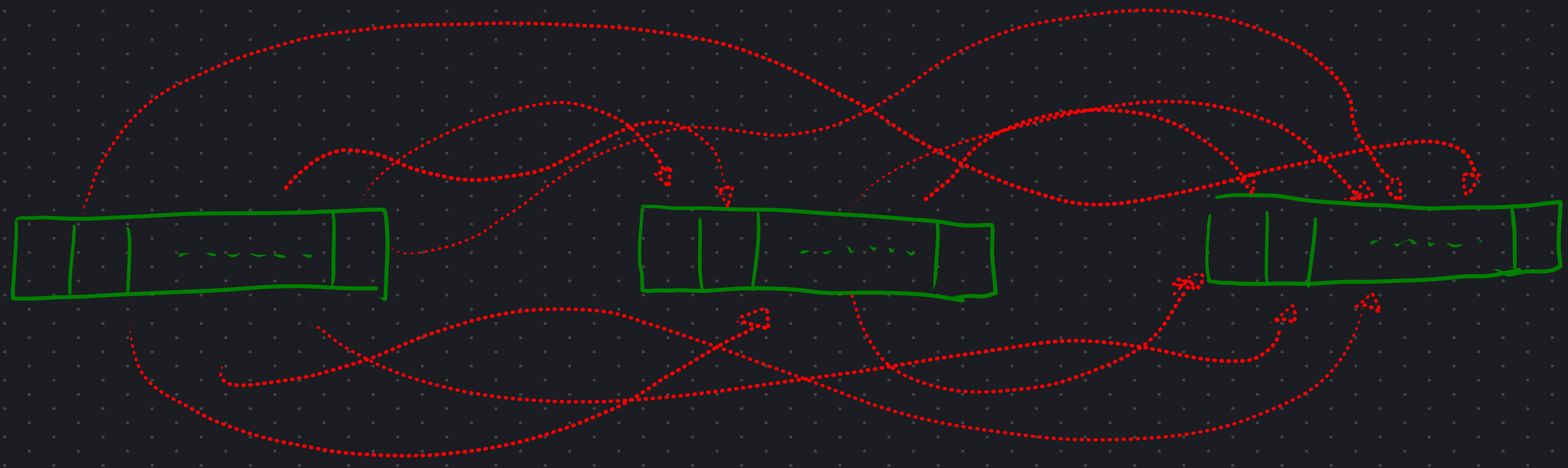
OK, but HOW ?



Wait, what about the MLP ?

Attention is All You Need

ATTENTION is All You Need

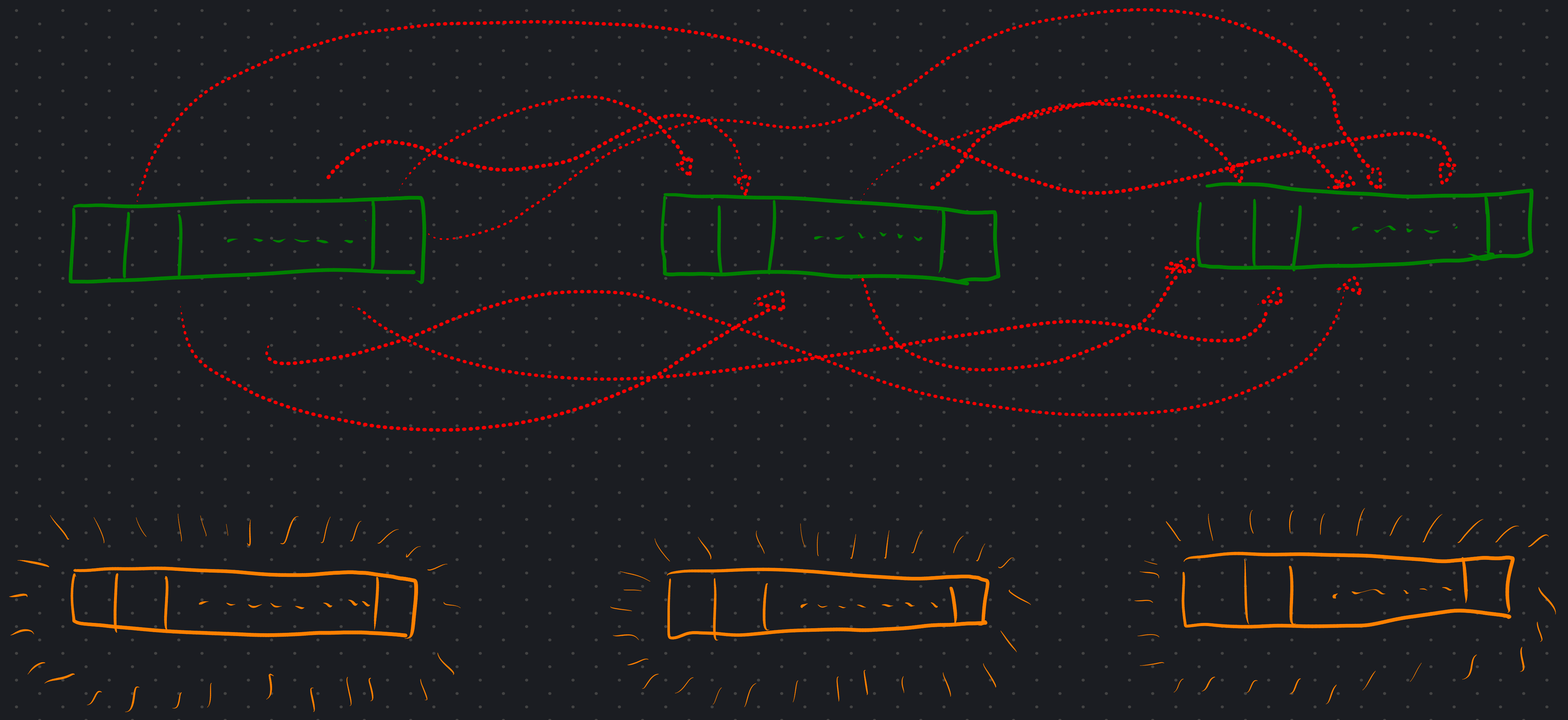


EMBEDDINGS

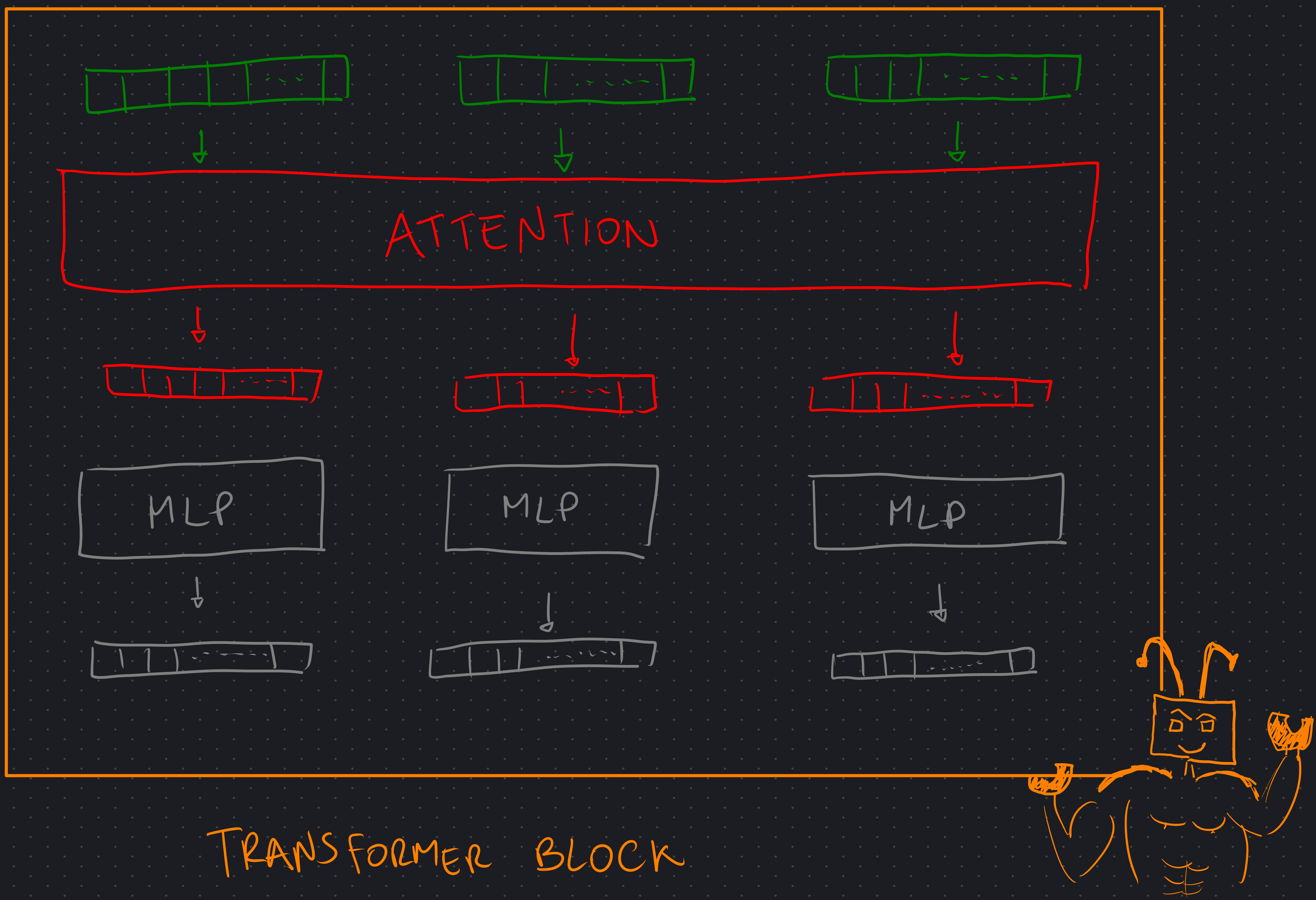
EXCHANGE INFORMATION

AND BECOME

ATTENTION is All You Need

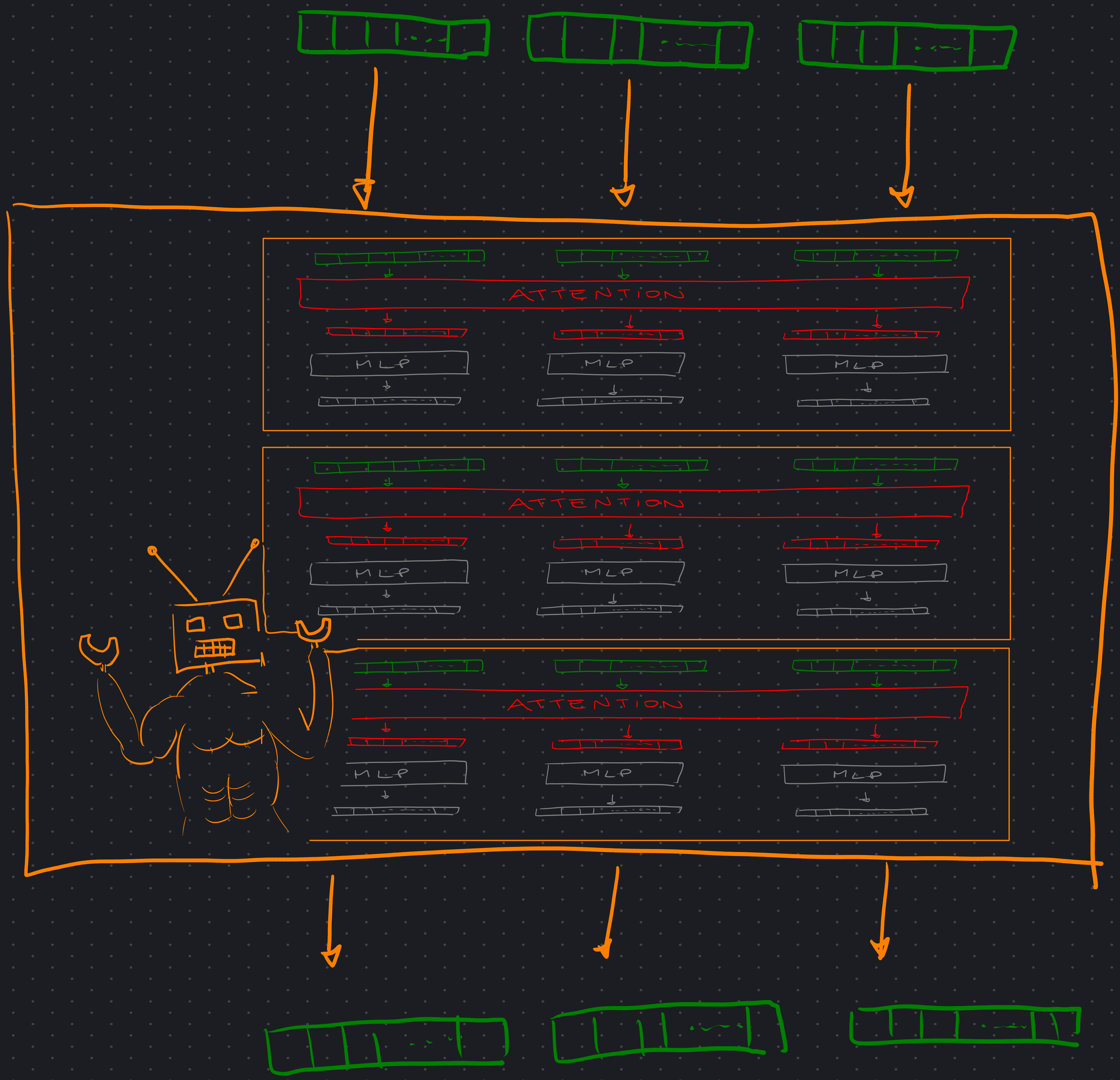


EMBEDDINGS
EXCHANGE INFORMATION
AND BECOME
NEW EMBEDDINGS!


ATTENTION
MLP
MLP
MLP
TRANSFORMER BLOCK

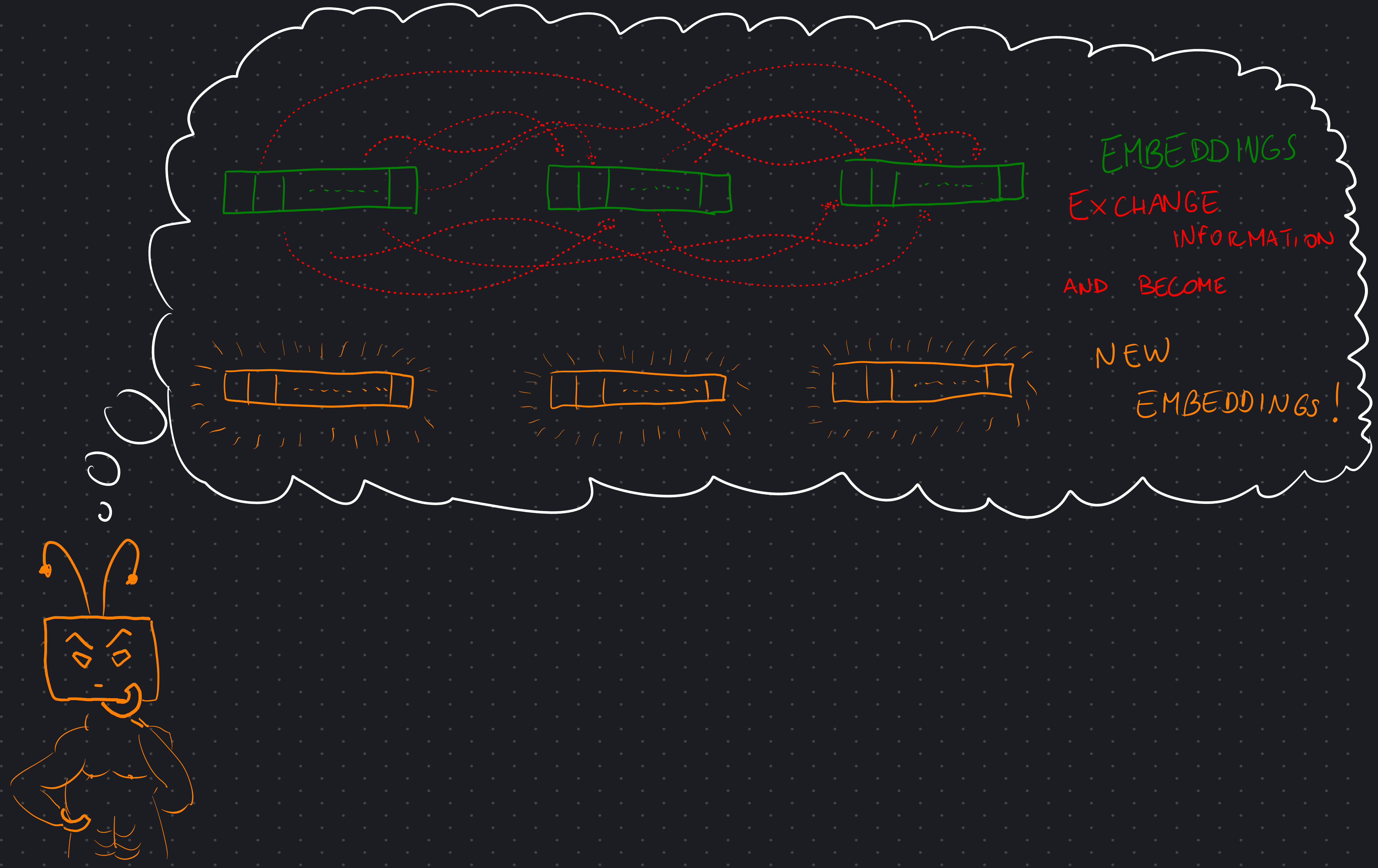
A TRANSFORMER is just a BUNCH of TRANSFORMER BLOCKS

(A.K.A: LAYERS)



EARLIER, we said:

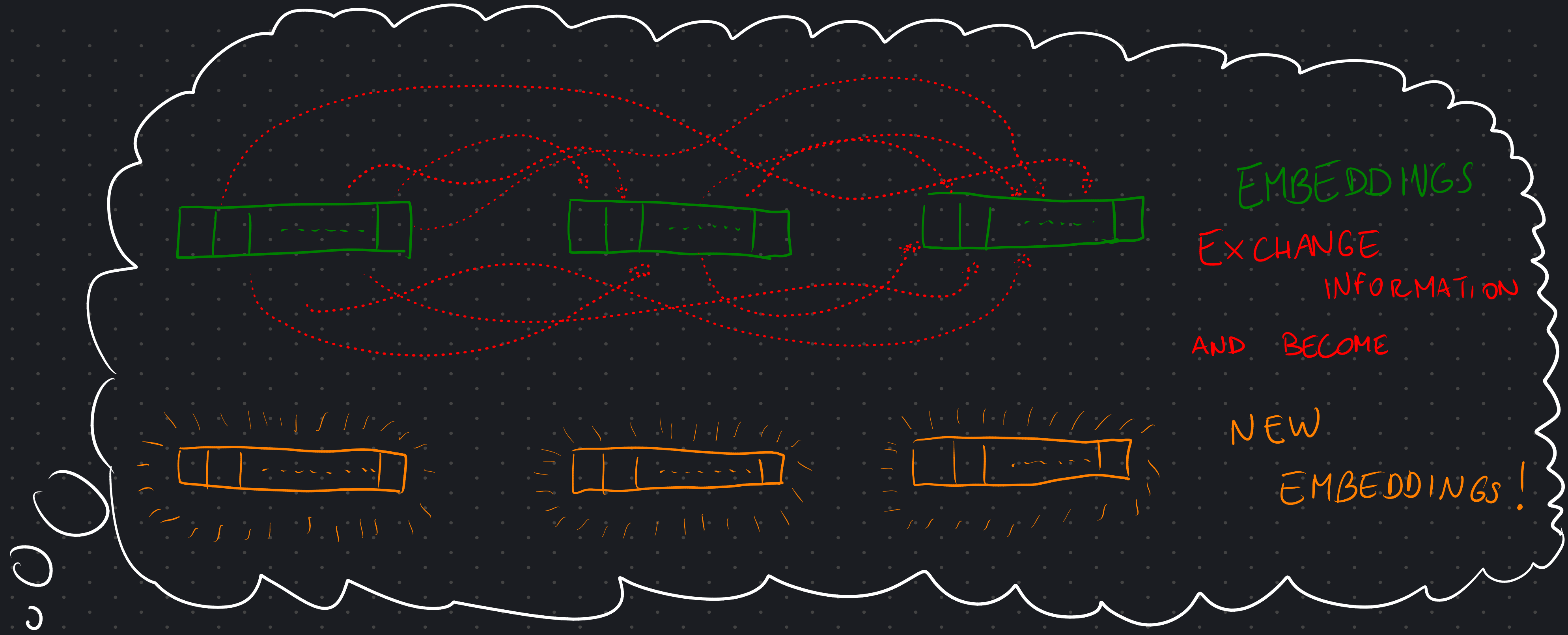
ATTENTION MAKES
EMBEDDINGS
EXCHANGE
INFORMATION



But in WHICH DIRECTION?

EARLIER, we said:

ATTENTION MAKES
EMBEDDINGS
EXCHANGE
INFORMATION



CAUSALITY

DECODER ✓

EARLIER, we said:

ATTENTION MAKES EMBEDDINGS EXCHANGE INFORMATION

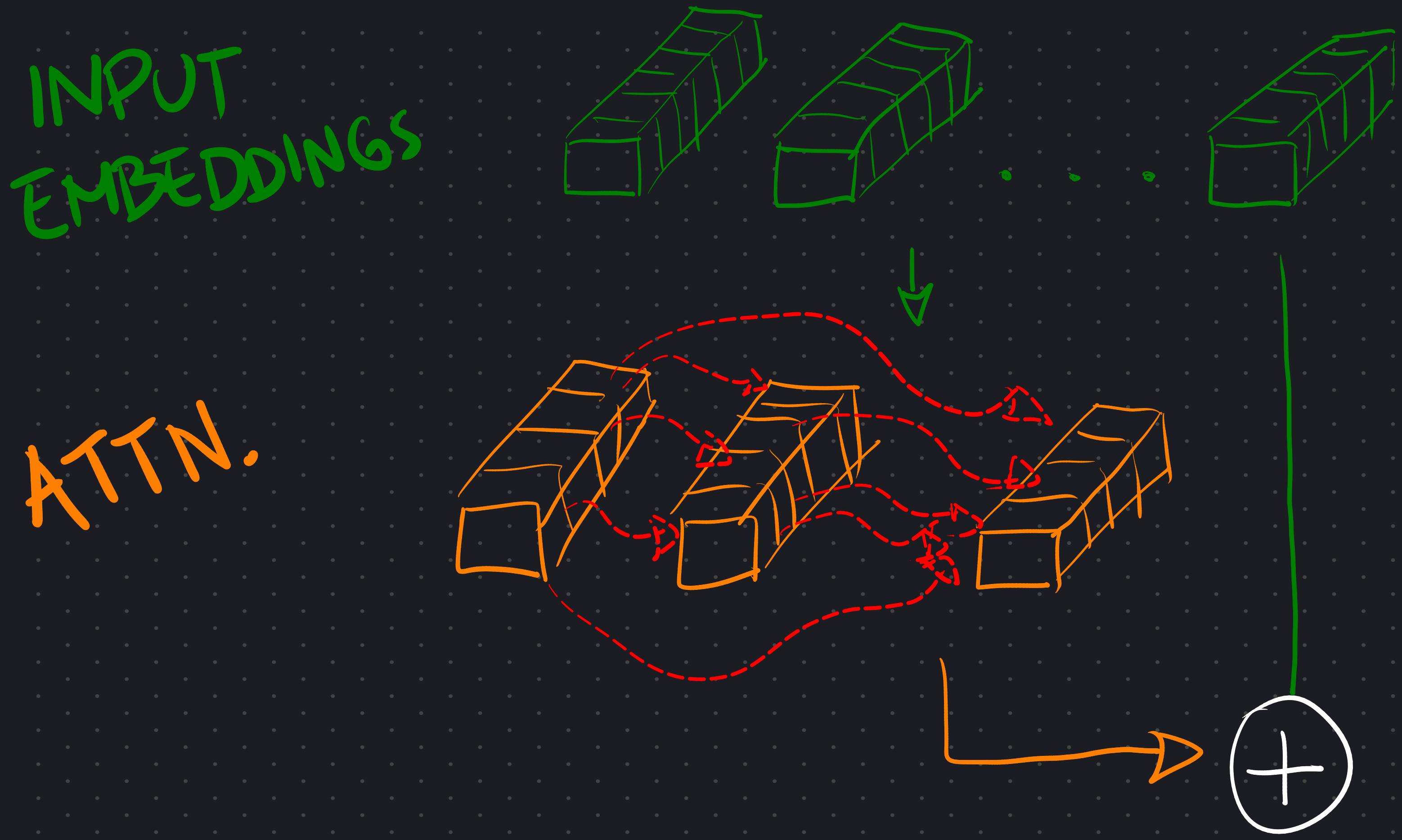
CAUSALITY

DECODER ✓

✗ ENCODER

2. THE RESIDUAL STREAM

INSIDE A TRANSFORMER BLOCK



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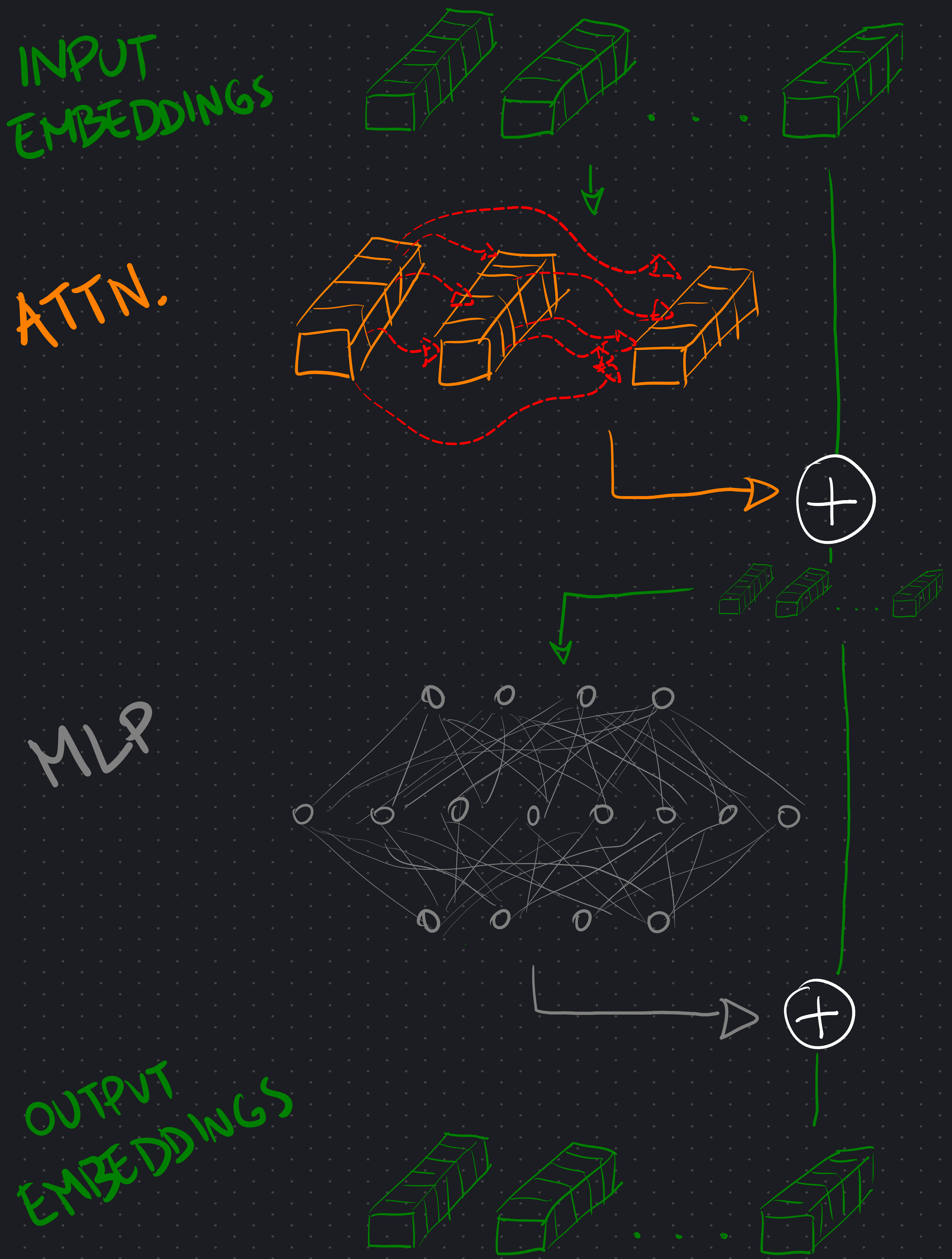
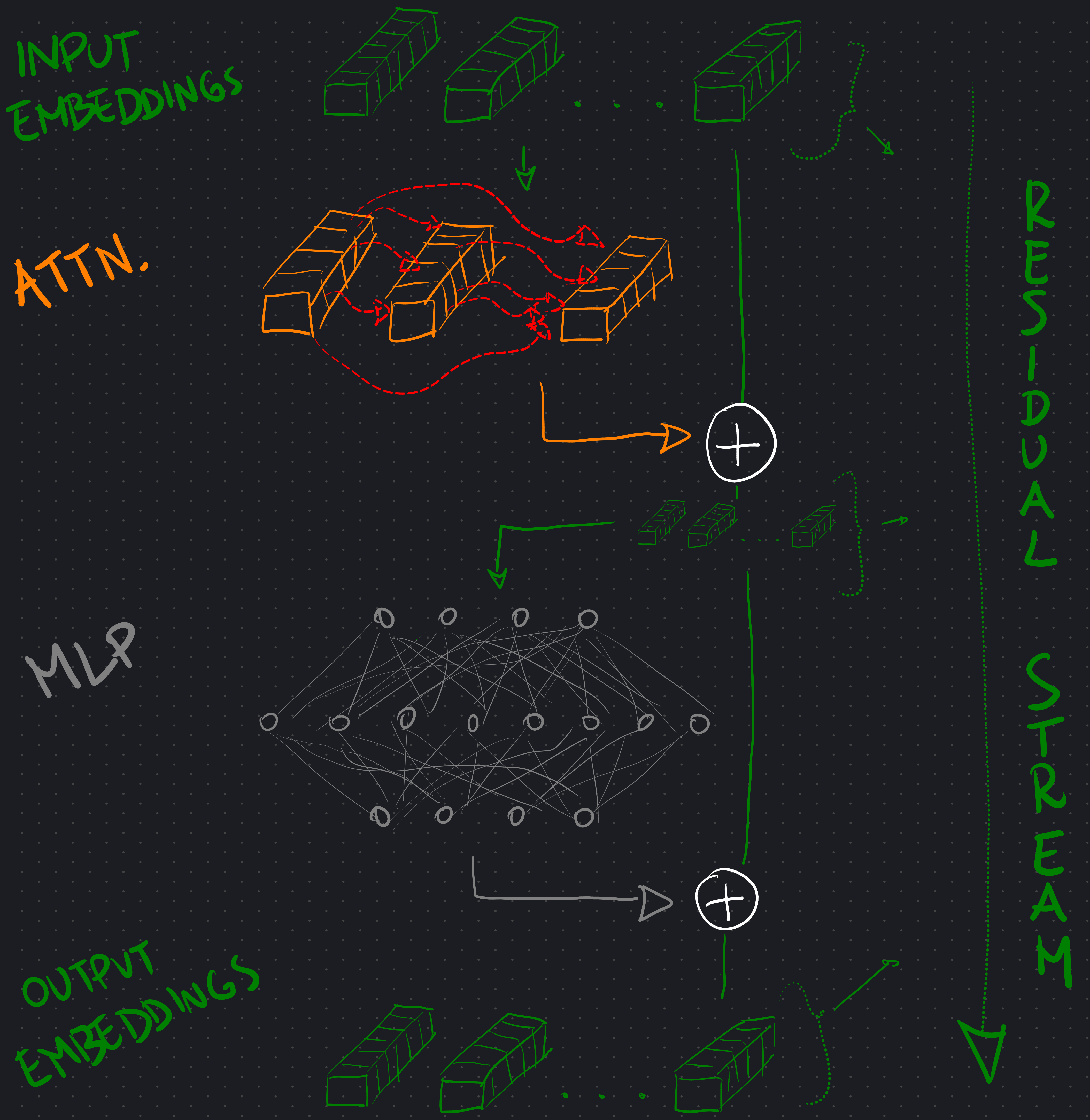
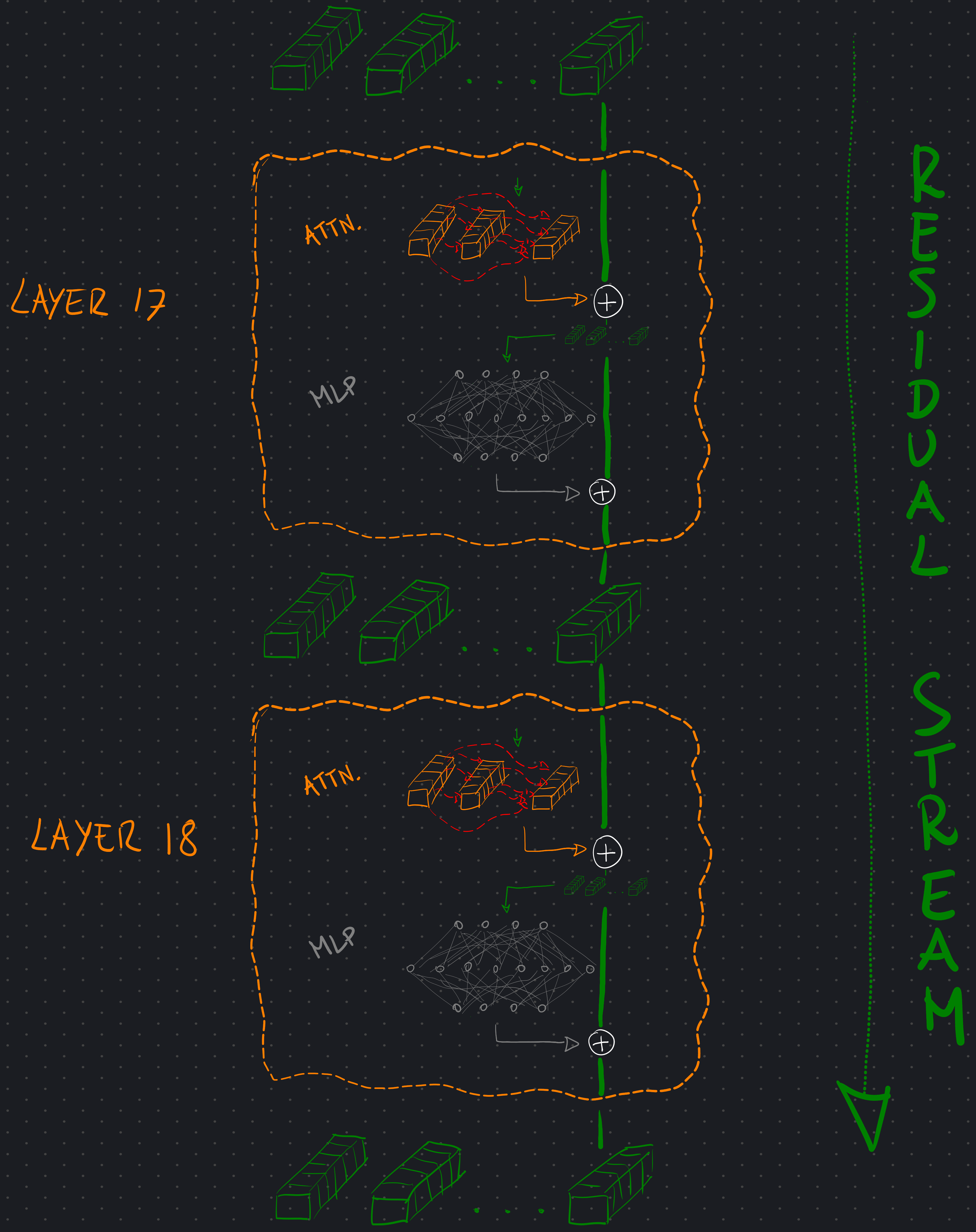
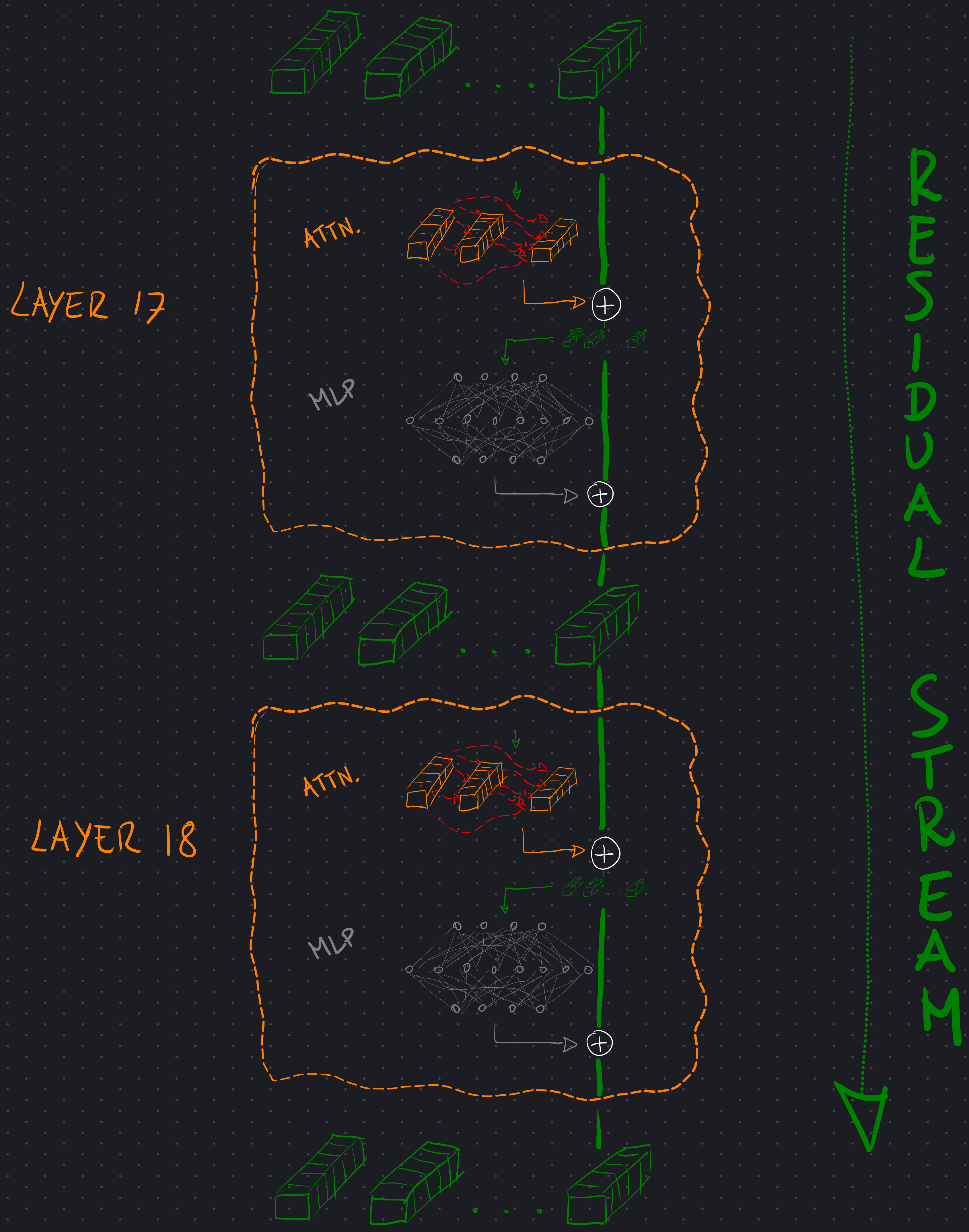


INSIDE A TRANSFORMER BLOCK



ACROSS MULTIPLE BLOCKS





ACROSS

MULTIPLE BLOCKS

THE **RESIDUAL STREAM** CONTAINS:

- FULL PROMPT
- THOUGHTS
- REASONING
- ANSWERS
- EVERYTHING!

We can think of EACH WEIGHT MATRIX as either READING FROM or WRITING TO the RESIDUAL STREAM

MLP_IN

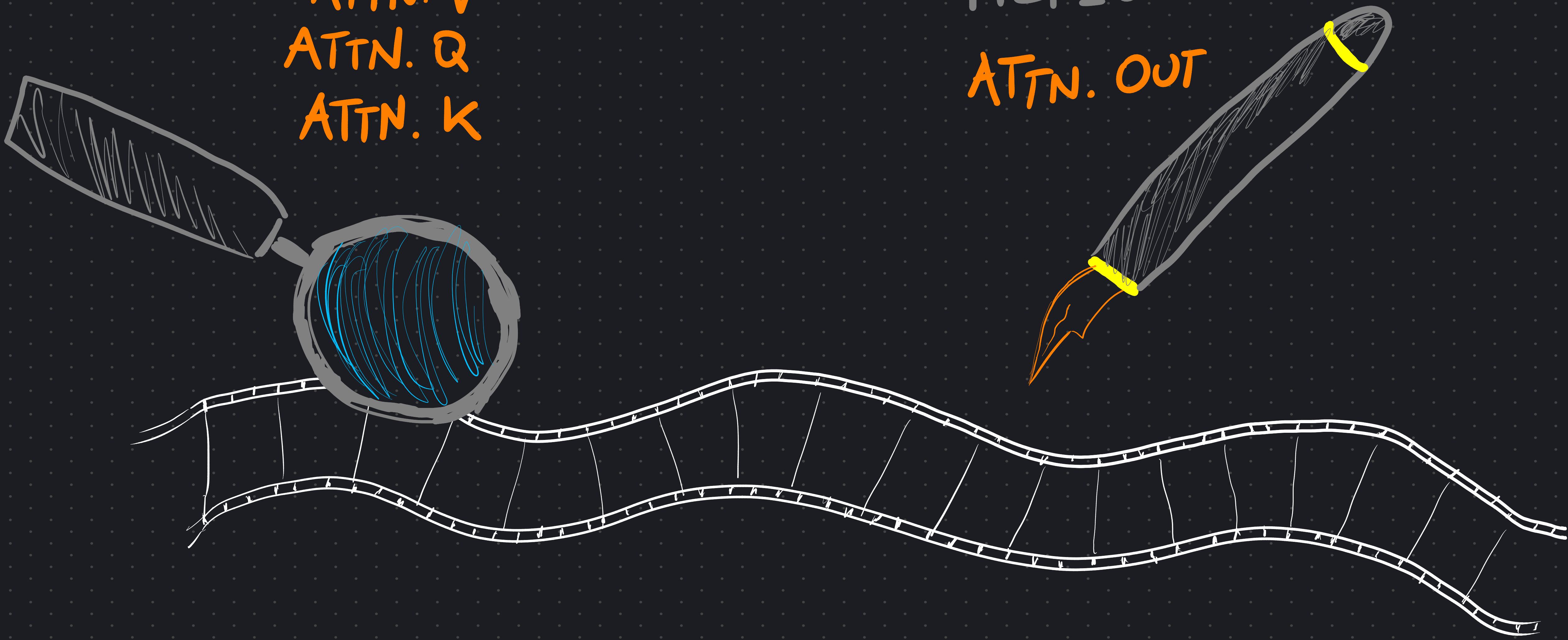
ATTN. V

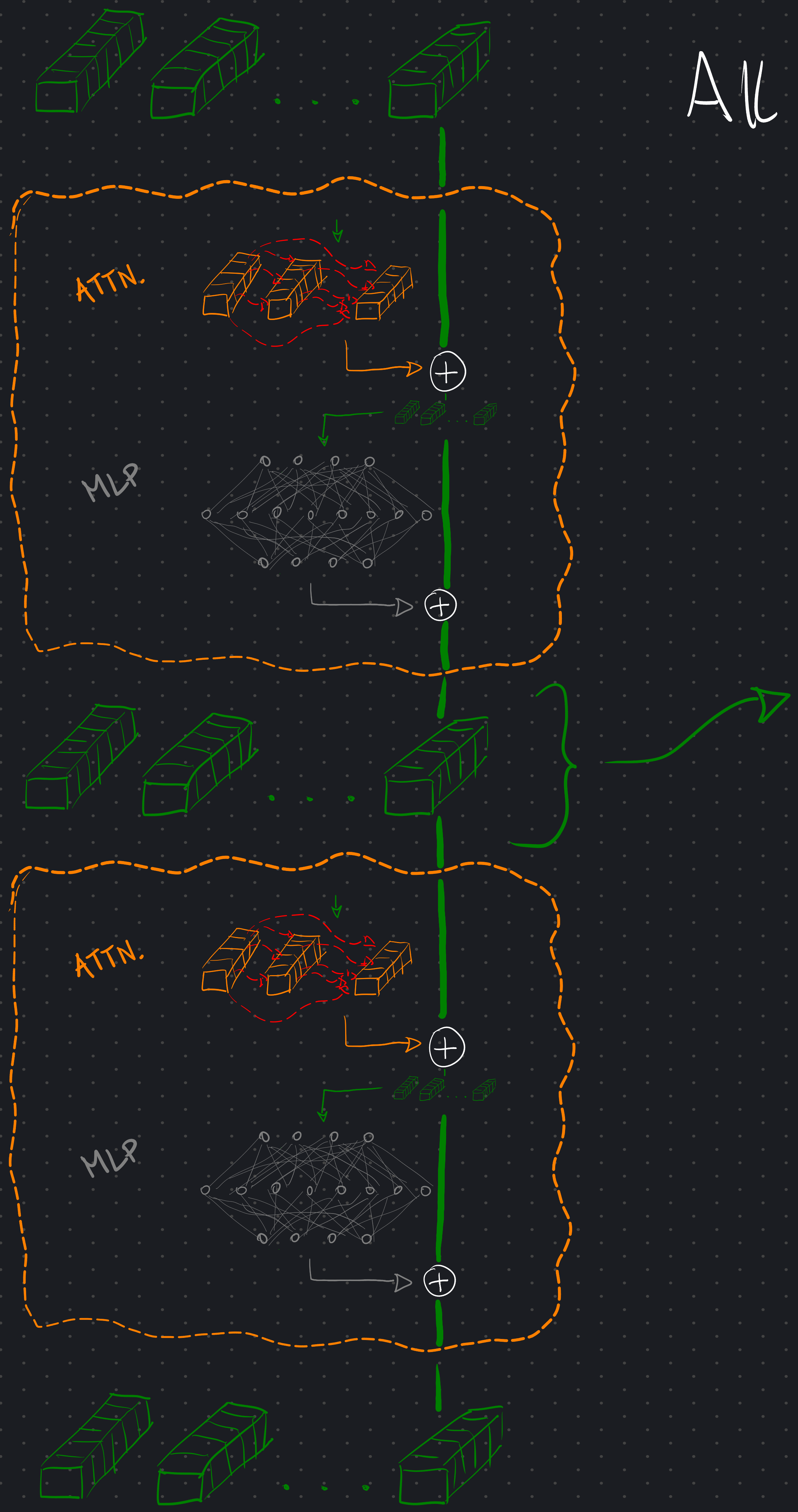
ATTN. Q

ATTN. K

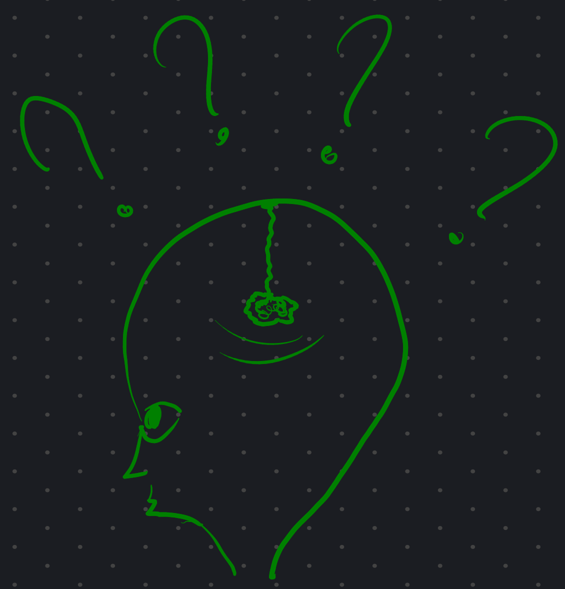
MLP_OUT

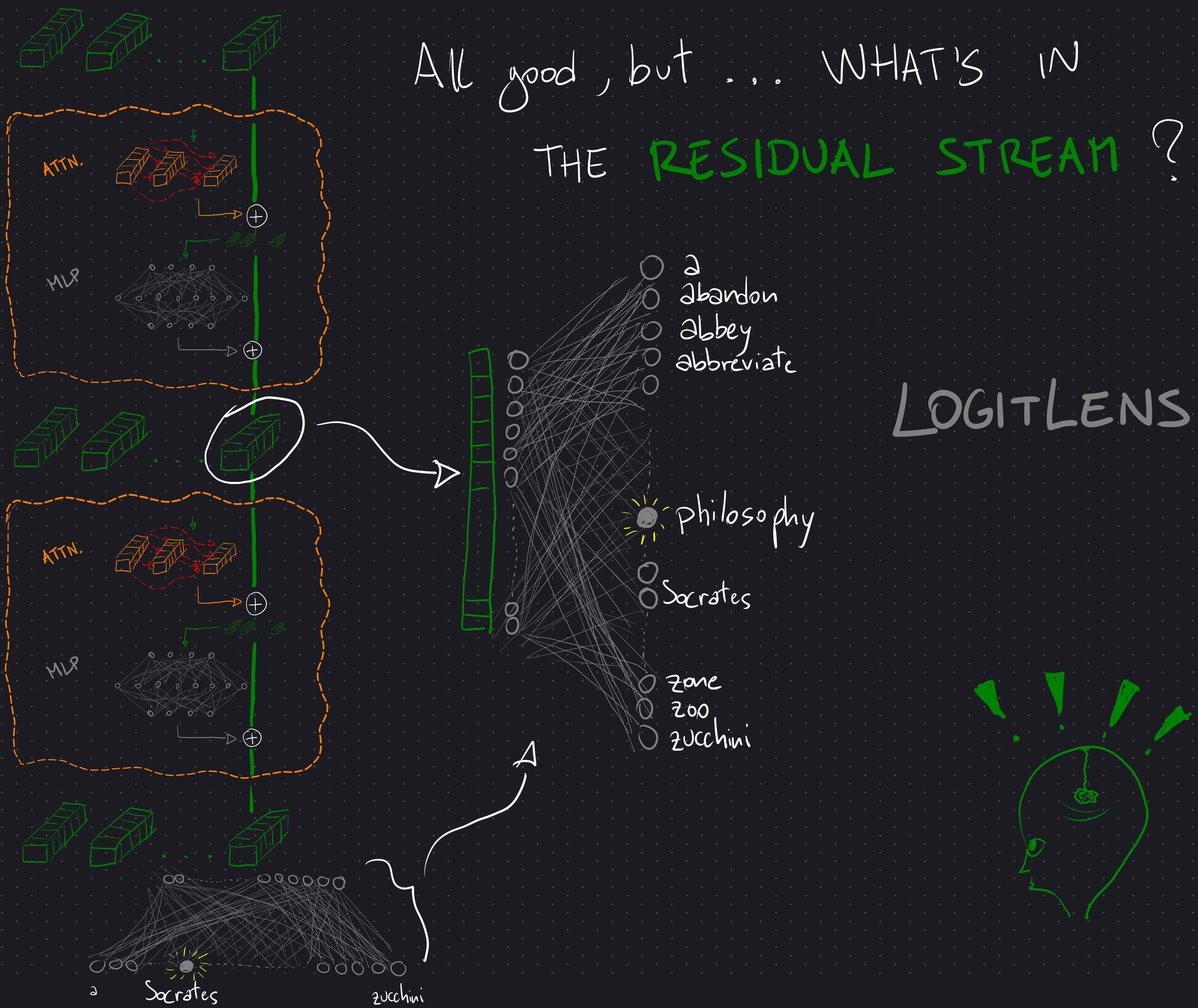
ATTN. OUT

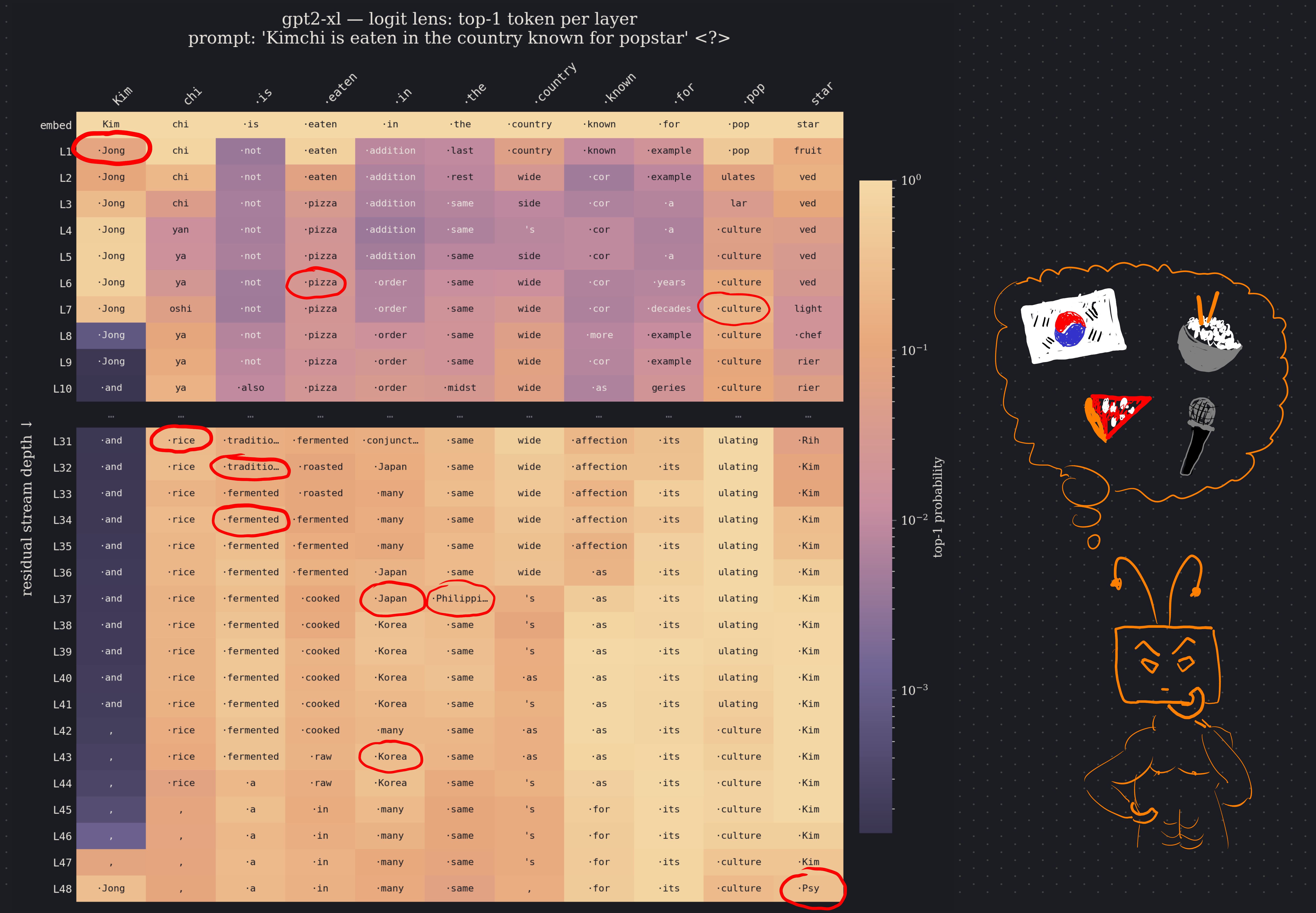




All good, but ... WHAT'S IN THE RESIDUAL STREAM ?




All good, but ... WHAT'S IN THE RESIDUAL STREAM ?
ATTN.
MLP
ATTN.
MLP
a
abandon
abbey
abbreviate
philosophy
Socrates
zone
zoo
zucchini
LOGITLENS
A
a
Socrates
zucchini


gpt2-xl — logit lens: top-1 token per layer
prompt: 'Kimchi is eaten in the country known for popstar' <?>
Kim chi ·is ·eaten ·in ·the ·country ·known ·for ·pop star
embed Kim chi ·is ·eaten ·in ·the ·country ·known ·for ·pop star
L1 ·Jong chi ·not ·eaten ·addition ·last ·country ·known ·example ·pop fruit
L2 ·Jong chi ·not ·eaten ·addition ·rest wide ·cor ·example ulates ved
L3 ·Jong chi ·not ·pizza ·addition ·same side ·cor ·a lar ved
L4 ·Jong yan ·not ·pizza ·addition ·same 's ·cor ·a ·culture ved
L5 ·Jong ya ·not ·pizza ·addition ·same side ·cor ·a ·culture ved
L6 ·Jong ya ·not ·pizza ·order ·same wide ·cor ·years ·culture ved
L7 ·Jong oshi ·not ·pizza ·order ·same wide ·cor ·decades ·culture light
L8 ·Jong ya ·not ·pizza ·order ·same wide ·more ·example ·culture ·chef
L9 ·Jong ya ·not ·pizza ·order ·same wide ·cor ·example ·culture rier
L10 ·and ya ·also ·pizza ·order ·midst wide ·as geries ·culture rier
… … … … … … … … … … …
L31 ·and ·rice ·traditio… ·fermented ·conjunct… ·same wide ·affection ·its ulating ·Rih
L32 ·and ·rice ·traditio… ·roasted ·Japan ·same wide ·affection ·its ulating ·Kim
L33 ·and ·rice ·fermented ·roasted ·many ·same wide ·affection ·its ulating ·Kim
L34 ·and ·rice ·fermented ·fermented ·many ·same wide ·affection ·its ulating ·Kim
L35 ·and ·rice ·fermented ·fermented ·many ·same wide ·affection ·its ulating ·Kim
L36 ·and ·rice ·fermented ·fermented ·Japan ·same wide ·as ·its ulating ·Kim
L37 ·and ·rice ·fermented ·cooked ·Japan ·Philippi… 's ·as ·its ulating ·Kim
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L41 ·and ·rice ·fermented ·cooked ·Korea ·same 's ·as ·its ulating ·Kim
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L43 , ·rice ·fermented ·raw ·Korea ·same ·as ·as ·its ·culture ·Kim
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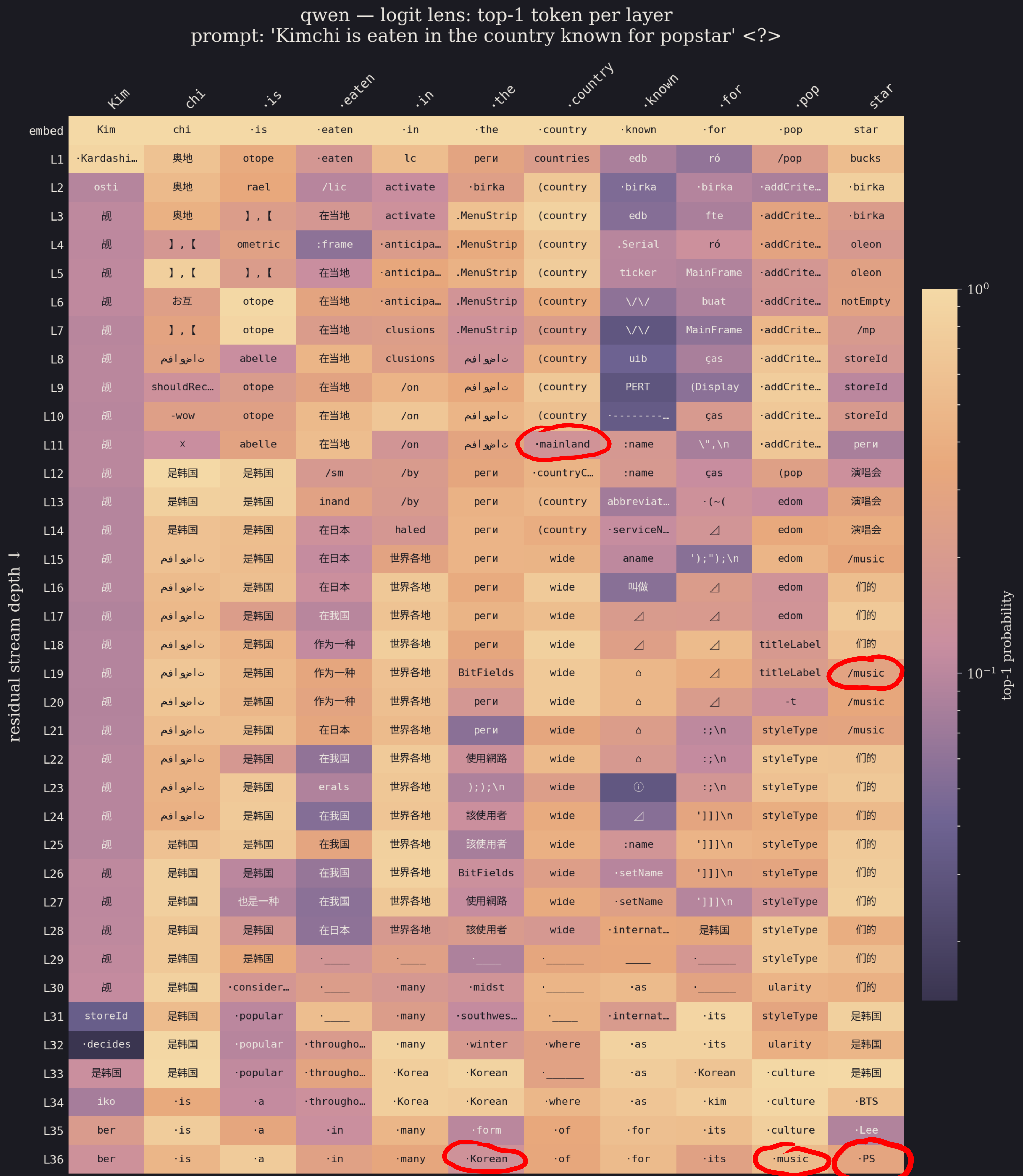
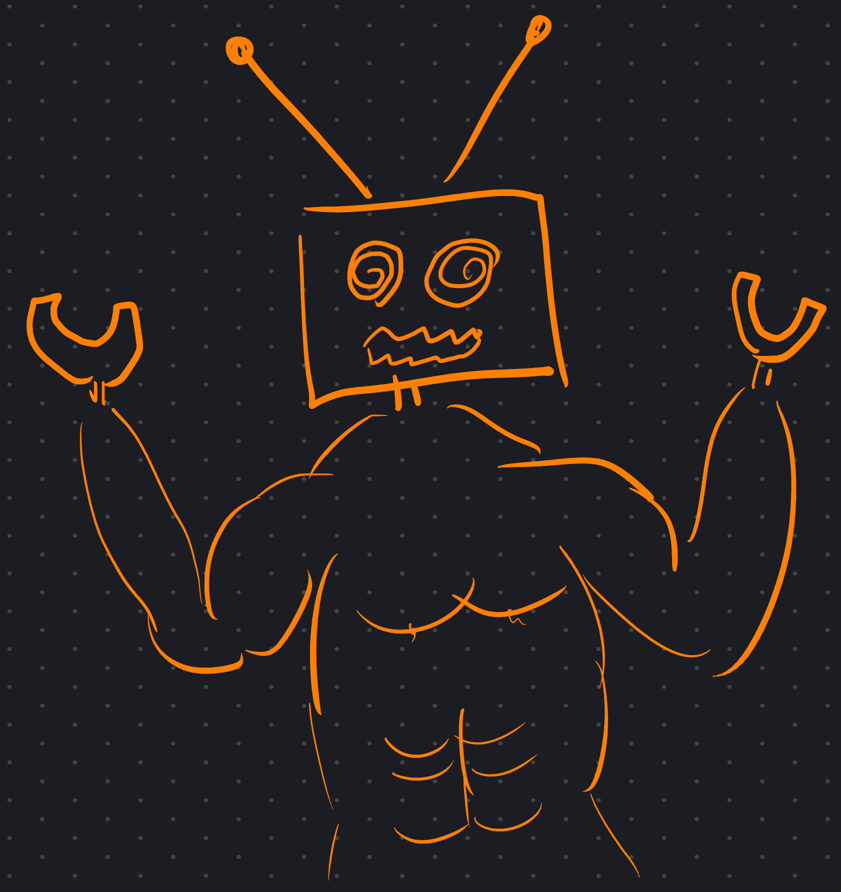
L45 , , ·a ·in ·many ·same 's ·for ·its ·culture ·Kim
L46 , , ·a ·in ·many ·same 's ·for ·its ·culture ·Kim
L47 , , ·a ·in ·many ·same 's ·for ·its ·culture ·Kim
L48 ·Jong , ·a ·in ·many ·same , ·for ·its ·culture ·Psy
residual stream depth ↓
10^0
10^{-1}
10^{-2}
10^{-3}
top-1 probability



Is this model
JUST CRAZY ?

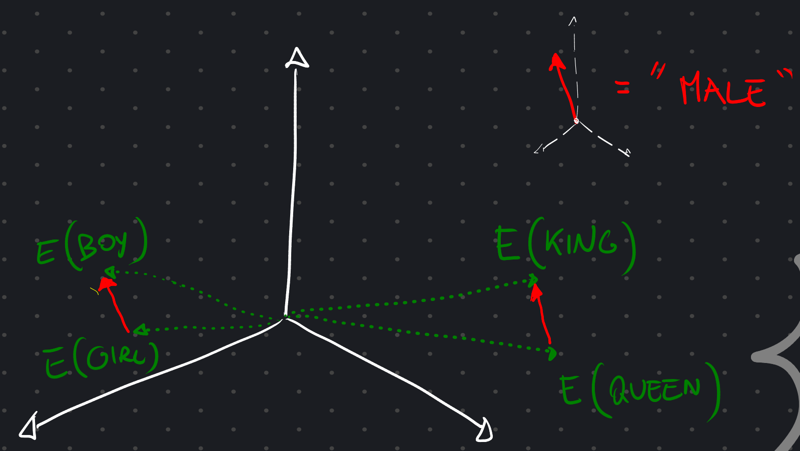


3. SUPERPOSITION

"Professor Oak's Words echoed: "

DIRECTIONS encode MEANING

$$E(\text{KING}) - E(\text{BOY}) + E(\text{GIRL}) \approx E(\text{QUEEN})$$



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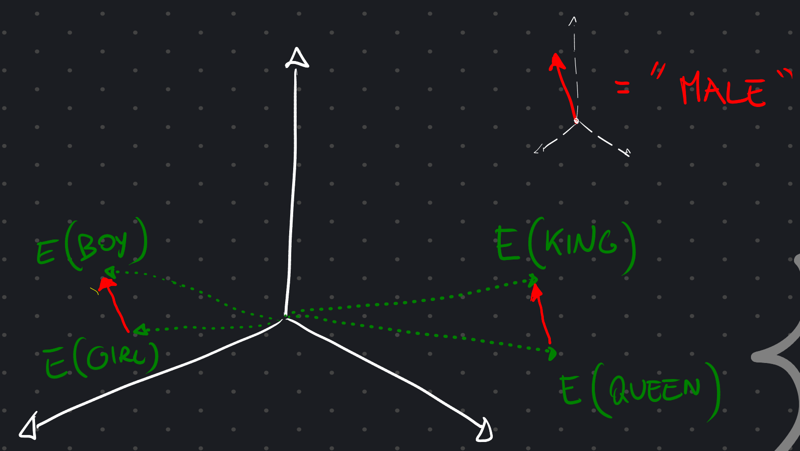
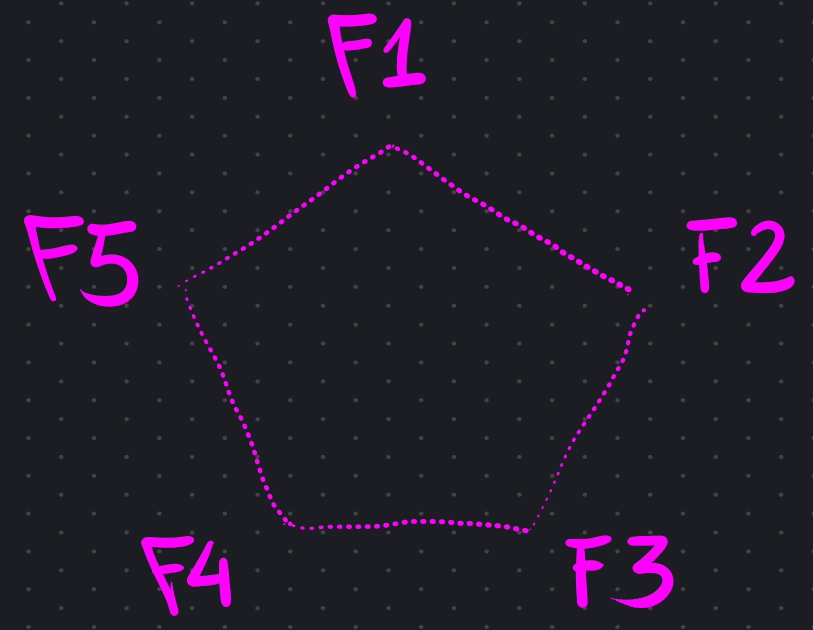
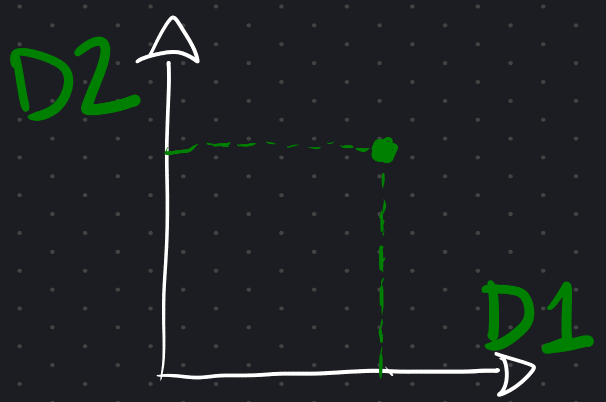


BUT GPT2-XL HAS A 1600-DIMENSIONAL RESIDUAL STREAM

HOW DOES IT FIT EVERYTHING IN THERE?

DATA HAS 5 FEATURES





MODEL HAS 2 DIMENSIONS

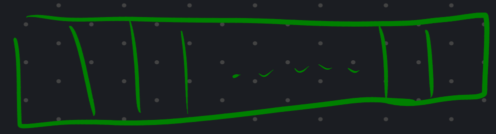
F1 $\rightsquigarrow$ D1

F2 $\rightsquigarrow$ D2

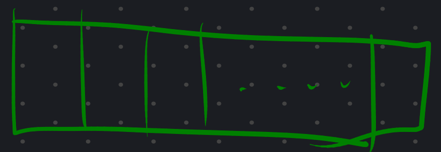
BUT WHAT IF FEATURES ARE SPARSE ?

F1 , F3

F2

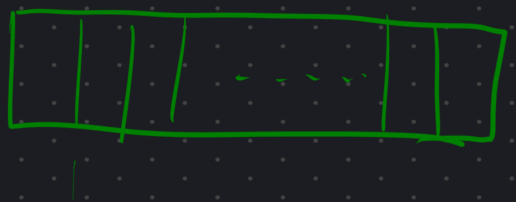


F3

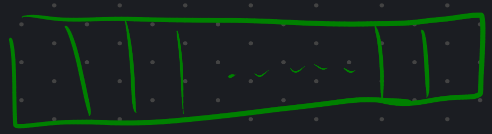


BUT WHAT IF FEATURES ARE SPARSE?

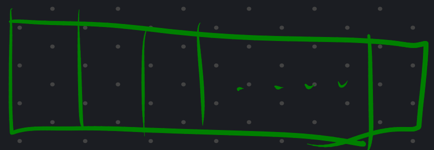
F1, F3

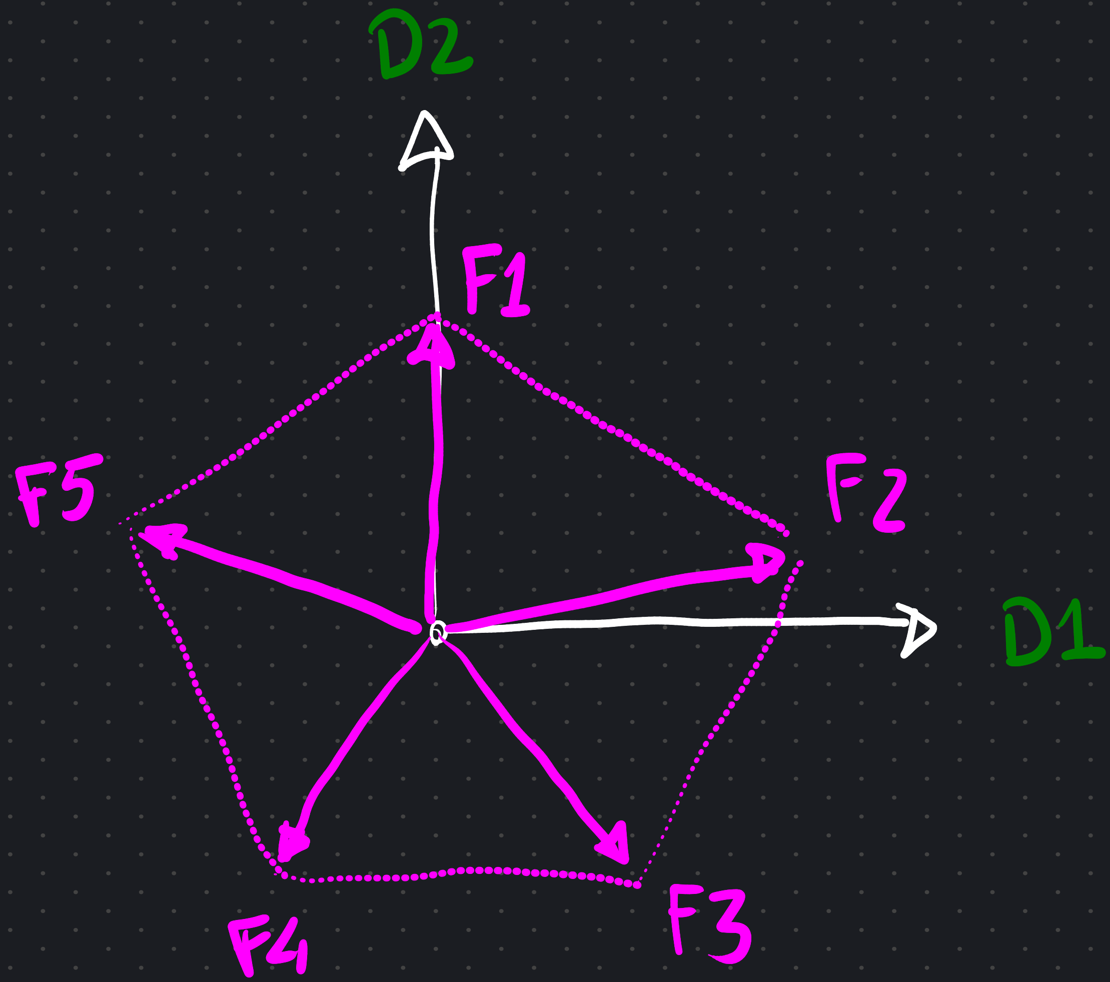


F2



F3





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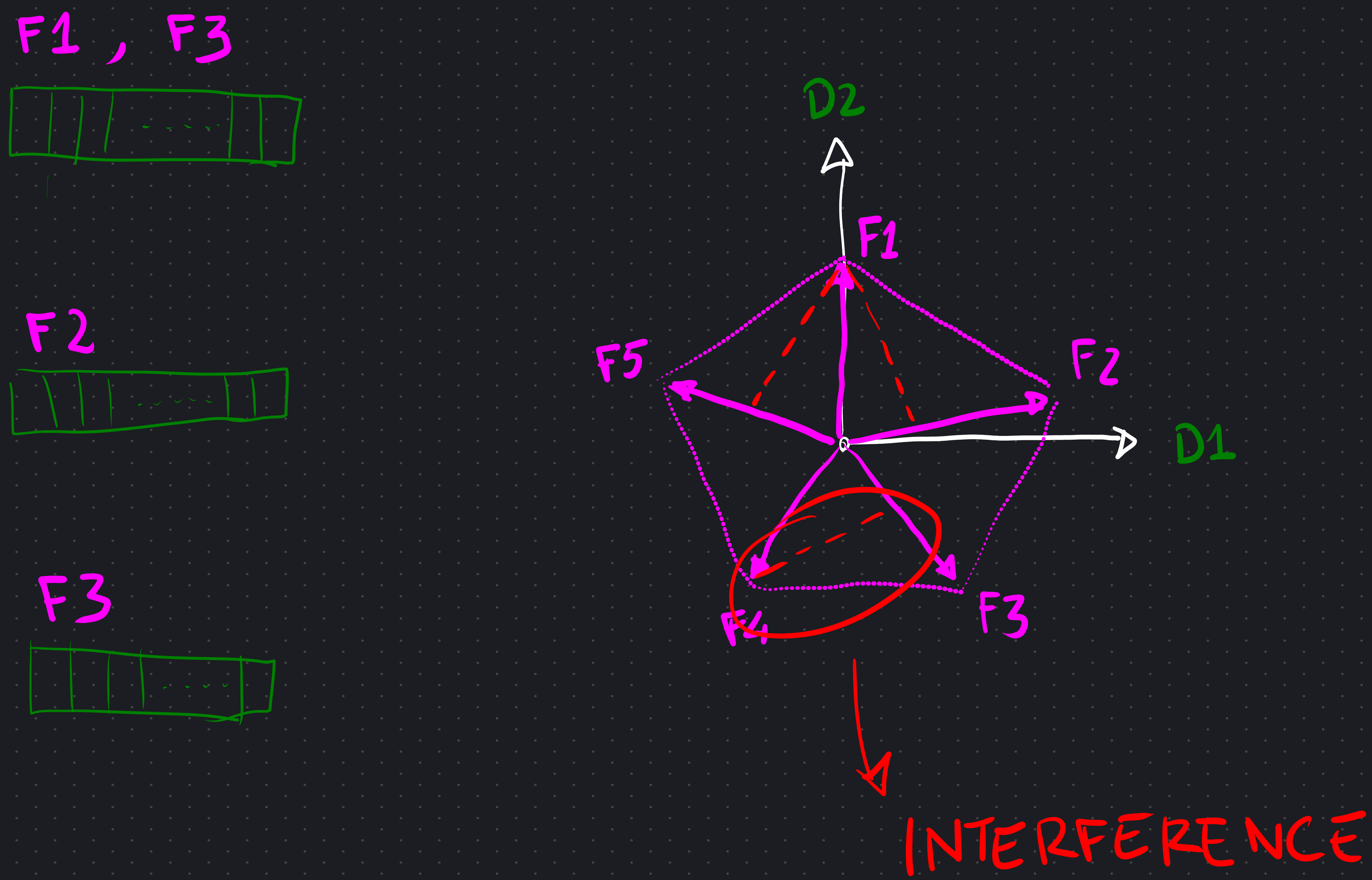
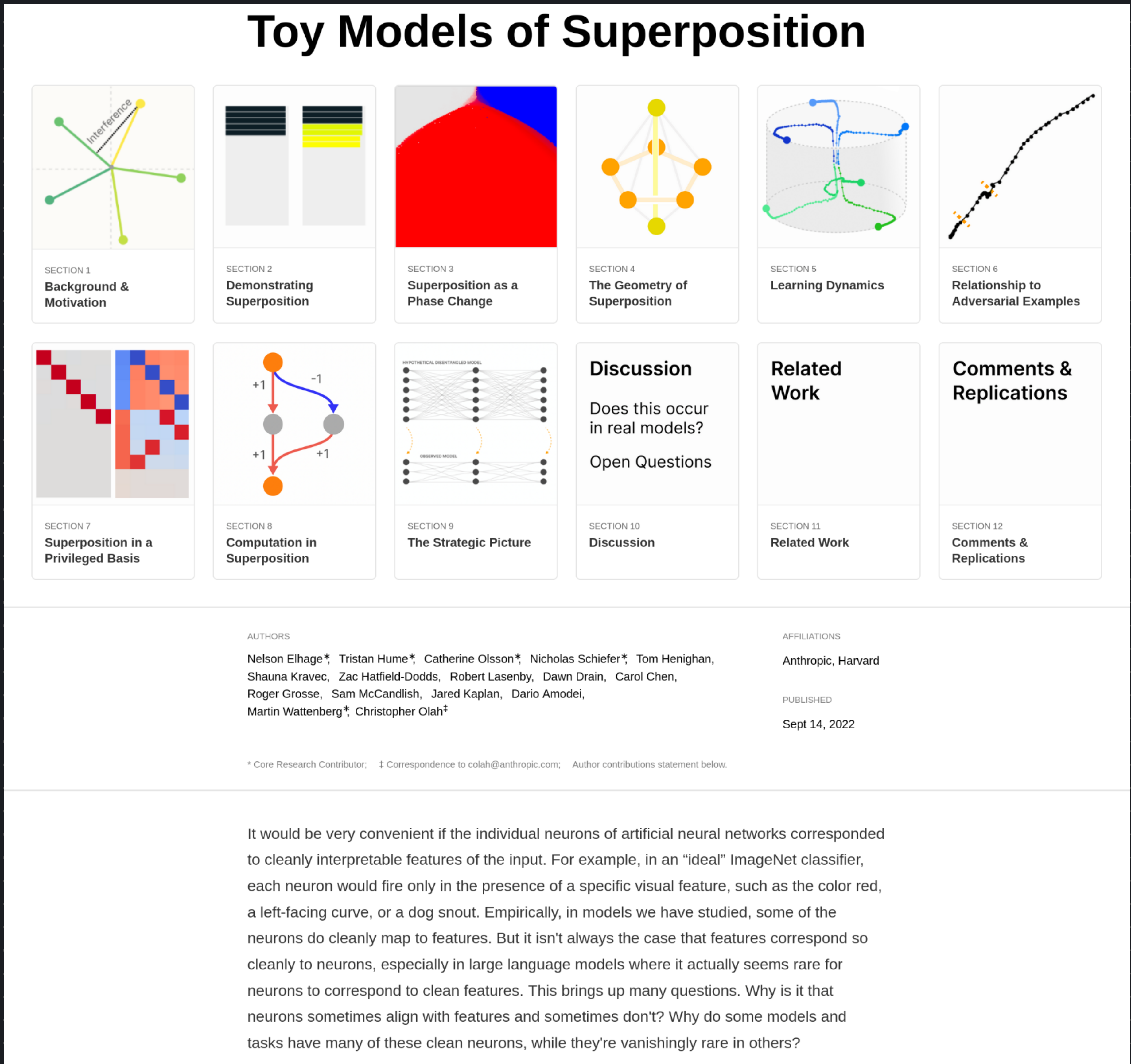


SOME NEURONS MAP TO MULTIPLE FEATURES



Toy Models of Superposition

SECTION 1
Background & Motivation

SECTION 2
Demonstrating Superposition

SECTION 3
Superposition as a Phase Change

SECTION 4
The Geometry of Superposition

SECTION 5
Learning Dynamics

SECTION 6
Relationship to Adversarial Examples

SECTION 7
Superposition in a Privileged Basis

SECTION 8
Computation in Superposition

SECTION 9
The Strategic Picture

Discussion
Does this occur in real models?
Open Questions

SECTION 10
Discussion

Related Work

SECTION 11
Related Work

Comments & Replications

SECTION 12
Comments & Replications

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It would be very convenient if the individual neurons of artificial neural networks corresponded to cleanly interpretable features of the input. For example, in an "ideal" ImageNet classifier, each neuron would fire only in the presence of a specific visual feature, such as the color red, a left-facing curve, or a dog snout. Empirically, in models we have studied, some of the neurons do cleanly map to features. But it isn't always the case that features correspond so cleanly to neurons, especially in large language models where it actually seems rare for neurons to correspond to clean features. This brings up many questions. Why is it that neurons sometimes align with features and sometimes don't? Why do some models and tasks have many of these clean neurons, while they're vanishingly rare in others?

This is called

SUPERPOSITION

- HOW DOES COMPUTATION IN SUPERPOSITION HAPPEN?
- HOW DO WE EXTRACT FEATURES FROM THE RESIDUAL STREAM?
- HOW DO WE DECYPHER THE RESIDUAL STREAM?
- WHAT'S INSIDE THE HIDDEN LAYER OF THE MLP?

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FOR THIS AND MUCH MORE SHIT I DON'T KNOW